

Q1: A uniformly charged disc of radius R having surface charge density σ is placed in the xy plane with its center at the origin. Find the electric field intensity along the z -axis at a distance Z from origin :-

(A)
$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$$

(B)
$$E = \frac{\sigma}{2\epsilon_0} \left(1 + \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$$

(C)
$$E = \frac{2\epsilon_0}{\sigma} \left(\frac{1}{(Z^2 + R^2)^{1/2}} + Z \right)$$

(D)
$$E = \frac{\sigma}{2\epsilon_0} \left(\frac{1}{(Z^2 + R^2)} + \frac{1}{Z^2} \right)$$

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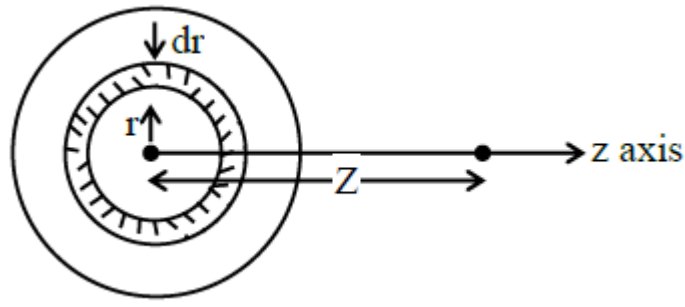
(B) $E = \frac{\sigma}{2\epsilon_0} \left(1 + \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$

(C) $E = \frac{2\epsilon_0}{\sigma} \left(\frac{1}{(Z^2 + R^2)^{1/2}} + Z \right)$

(D) $E = \frac{\sigma}{2\epsilon_0} \left(\frac{1}{(Z^2 + R^2)} + \frac{1}{Z^2} \right)$

Solution:

Consider a small ring of radius r and thickness dr on disc.



area of elemental ring on disc

$$dA = 2\pi r dr$$

charge on this ring $dq = \sigma dA$

$$dE_z = \frac{k(dq)z}{(z^2 + r^2)^{3/2}}$$

$$E = \int_0^R dE_z = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{R^2 + z^2}} \right]$$

Q2: There are 10^{10} radioactive nuclei in a given radioactive element, Its half-life time is 1 minute. How many nuclei will remain after 30 seconds ?

$(\sqrt{2} = 1.414)$

(A) 2×10^{10}

(B) 7×10^9

(C) 10^5

(D) 4×10^{10}

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Solution:

$$\begin{aligned}\frac{N}{N_0} &= \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} \\ \frac{N}{10^{10}} &= \left(\frac{1}{2}\right)^{\frac{30}{60}} \\ \Rightarrow N &= 10^{10} \times \left(\frac{1}{2}\right)^{\frac{1}{2}} = \frac{10^{10}}{\sqrt{2}} \approx 7 \times 10^9\end{aligned}$$

Q3: Which of the following is not a dimensionless quantity?

- (A) Relative magnetic permeability (μ_r)
- (B) Power factor
- (C) Permeability of free space (μ_0)
- (D) Quality factor

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Solution:

$$[\mu_r] = 1 \text{ as } \mu_r = \frac{\mu}{\mu_m}$$

$$[\text{power factor } (\cos \phi)] = 1$$

$$\mu_0 = \frac{B_0}{H} \text{ (unit= } NA^{-2}) : \text{ Not dimensionless}$$

$$[\mu_0] = [MLT^{-2}A^{-2}]$$

$$\text{quality factor } (Q) = \frac{\text{Energy stored}}{\text{Energy dissipated per cycle}}$$

So Q is unitless & dimensionless.

Q4: If E and H represents the intensity of electric field and magnetising field respectively, then the unit of E/H will be :

- (A) ohm
- (B) mho
- (C) joule
- (D) newton

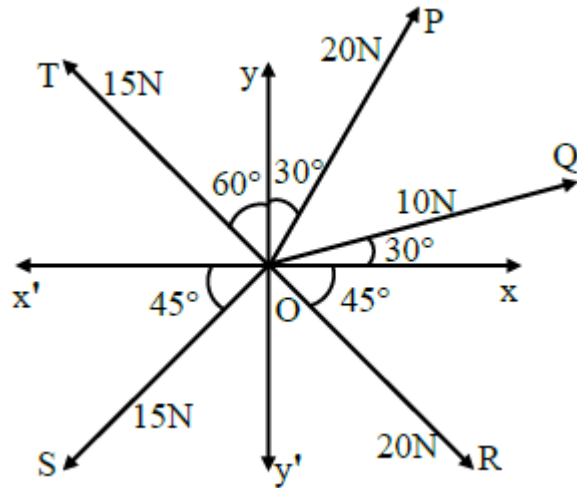
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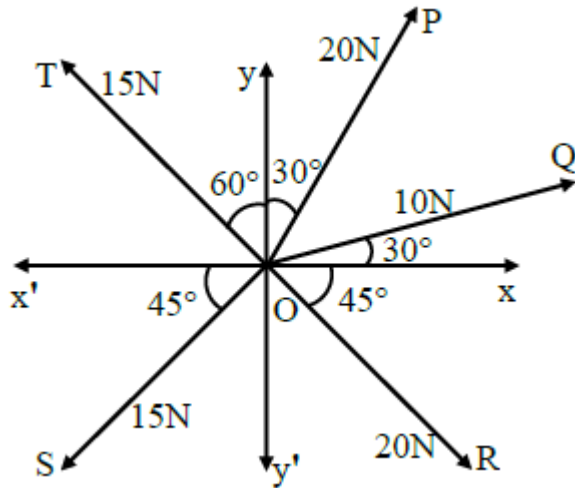
$$\text{Unit of } \frac{E}{H} \text{ is } \frac{\text{volt/metre}}{\text{Ampere/metre}} = \frac{\text{volt}}{\text{Ampere}} = \text{ohm}$$

Q5: The resultant of these forces $\vec{OP}$, $\vec{OQ}$, $\vec{OR}$, $\vec{OS}$ and $\vec{OT}$ is approximately N.
 [Take $\sqrt{3} = 1.7$, $\sqrt{2} = 1.4$ Given $\hat{i}$ and $\hat{j}$ unit vectors along x, y axis]



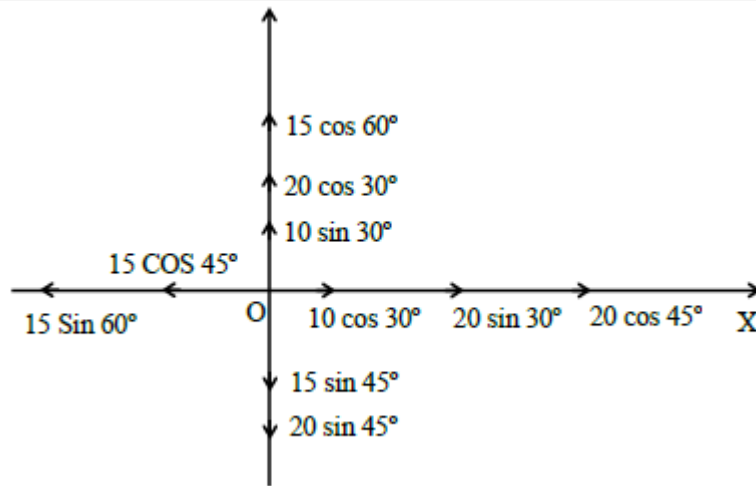
- (A) $9.25\hat{i} + 5\hat{j}$
- (B) $3\hat{i} + 15\hat{j}$
- (C) $2.5\hat{i} - 14.5\hat{j}$
- (D) $-1.5\hat{i} - 15.5\hat{j}$

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Solution:



$$\vec{F}_x = \left(10 \times \frac{\sqrt{3}}{2} + 20 \left(\frac{1}{2} \right) + 20 \left(\frac{1}{\sqrt{2}} \right) - 15 \left(\frac{1}{\sqrt{2}} \right) - 15 \left(\frac{\sqrt{3}}{2} \right) \right) \hat{i}$$

$$= 9.25 \hat{i}$$

$$\vec{F}_y = \left(15 \left(\frac{1}{2} \right) + 20 \left(\frac{\sqrt{3}}{2} \right) + 10 \left(\frac{1}{2} \right) - 15 \left(\frac{1}{\sqrt{2}} \right) - 20 \left(\frac{1}{\sqrt{2}} \right) \right) \hat{j}$$

$$= 5 \hat{j}$$

Q6: A balloon carries a total load of 185 kg at normal pressure and temperature of 27°C . What load will the balloon carry on rising to a height at which the barometric pressure is 45 cm of Hg and the temperature is -7°C . Assuming the volume constant ?

- (A) 181.46 kg
- (B) 214.15 kg
- (C) 219.07 kg
- (D) 123.54 kg

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Solution:

$$\begin{aligned}
 P_m &= \rho RT \\
 \therefore \frac{P_1}{P_2} &= \frac{\rho_1 T_1}{\rho_2 T_2} \\
 \frac{\rho_1}{\rho_2} \Rightarrow \frac{P_1 T_2}{P_2 T_1} &= \left(\frac{76}{45} \right) \times \frac{266}{300} \\
 \frac{\rho_1}{\rho_2} \Rightarrow \frac{M_1}{M_2} &= \frac{76 \times 266}{45 \times 300} \\
 \therefore M_2 \Rightarrow \frac{45 \times 300 \times 185}{76 \times 266} &= 123.54 \text{ kg}
 \end{aligned}$$

Q7: An object is placed beyond the centre of curvature C of the given concave mirror. If the distance of the object is d_1 from C and the distance of the image formed is d_2 from C, the radius of curvature of this mirror is :

(A) $\frac{2d_1 d_2}{d_1 - d_2}$

(B) $\frac{2d_1 d_2}{d_1 + d_2}$

(C) $\frac{d_1 d_2}{d_1 + d_2}$

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(D) $\frac{d_1 d_2}{d_1 - d_2}$

Solution:

Using Newton's formula

$$(f + d_1)(f - d_2) = f^2$$

$$f^2 + fd_1 - fd_2 - d_1 d_2 = f^2$$

$$f = \frac{d_1 d_2}{d_1 - d_2}$$

$$\therefore R = \frac{2d_1 d_2}{d_1 - d_2}$$

Q8: A huge circular arc of length 4.4 ly subtends an angle '4s' at the centre of the circle. How long it would take for a body to complete 4 revolution if its speed is 8 AU per second ?

Given : $1 \text{ ly} = 9.46 \times 10^{15} \text{ m}$

$$1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$$

(A) $4.1 \times 10^8 \text{ s}$

(B) $4.5 \times 10^{10} \text{ s}$

(C) $3.5 \times 10^6 \text{ s}$

(D) $7.2 \times 10^8 \text{ s}$

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Solution:

$$R = \frac{\ell}{\theta}$$

$$\text{Time} = \frac{4 \times 2\pi R}{v} = \frac{4 \times 2\pi}{v} \left(\frac{\ell}{\theta} \right)$$

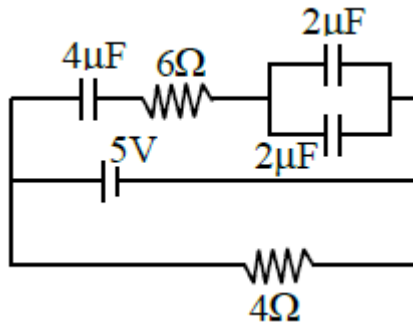
$$\text{put } \ell = 4.4 \times 9.46 \times 10^{15}$$

$$v = 8 \times 1.5 \times 10^{11}$$

$$\theta = \frac{4}{3600} \times \frac{\pi}{180} \text{ rad.}$$

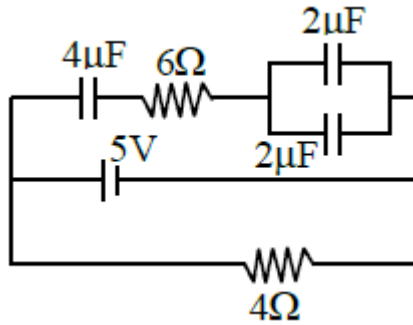
$$\text{we get time} = 4.5 \times 10^{10} \text{ sec}$$

Q9: Calculate the amount of charge on capacitor of $4\ \mu F$. The internal resistance of battery is $1\ \Omega$:



- (A) $8\ \mu C$
- (B) zero
- (C) $16\ \mu C$
- (D) $4\ \mu C$

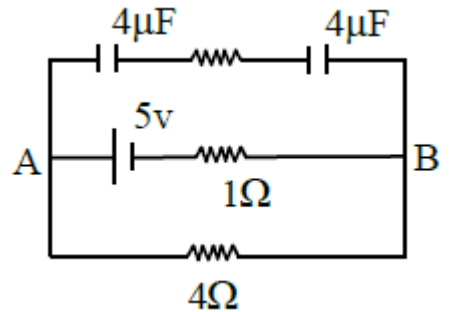
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Solution:

On simplifying circuit we get



No current in upper section of circuit.

$$\therefore V_{AB} = \frac{5}{4+1} \times 4 = 4\text{V}.$$

$$\therefore \theta = (C_{eq})v$$

$$\Rightarrow 2 \times 4 = 8\mu\text{C}$$

Q10: Moment of inertia of a square plate of side l about the axis passing through one of the corner and perpendicular to the plane of square plate is given by :

(A) $\frac{Ml^2}{6}$

(B) Ml^2

(C) $\frac{Ml^2}{12}$

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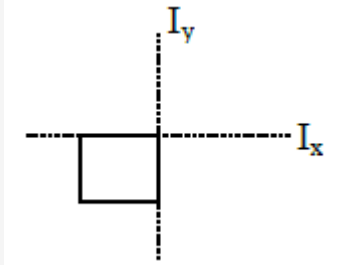
(B) Ml^2

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Solution:

According to perpendicular Axis theorem.



$$\begin{aligned} I_x + I_y &= I_z \\ I_z &\Rightarrow \frac{ml^2}{3} + \frac{ml^2}{3} \\ &= \frac{2ml^2}{3} \end{aligned}$$

Q11: For a transistor in CE mode to be used as an amplifier, it must be operated in :

- (A) Both cut-off and Saturation
- (B) Saturation region only
- (C) Cut-off region only
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Solution:

Active region of the CE transistor is a linear region and is best suited for its use as an amplifier.

Q12: An ideal gas is expanding such that $PT^3 = \text{constant}$. The coefficient of volume expansion of the gas is :

(A) $\frac{1}{T}$

(B) $\frac{2}{T}$

(C) $\frac{4}{T}$

(D) $\frac{3}{T}$

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Solution:

$$PT^3 = \text{constant}$$

$$\left(\frac{nRT}{V}\right) T^3 = \text{constant}$$

$$T^4 V^{-1} = \text{constant}$$

$$T^4 = kV$$

$$\Rightarrow 4 \frac{\Delta T}{T} = \frac{\Delta V}{V} \dots\dots (1)$$

$$\Delta V = V\gamma\Delta T \dots\dots\dots(2)$$

comparing (1) and (2)

we get

$$\gamma = \frac{4}{T}$$

Q13: In a photoelectric experiment, increasing the intensity of incident light :

- (A) increases the number of photons incident and also increases the K.E. of the ejected electrons
- (B) increases the frequency of photons incident and increases the K.E. of the ejected electrons.
- (C) increases the frequency of photons incident and the K.E. of the ejected electrons remains unchanged
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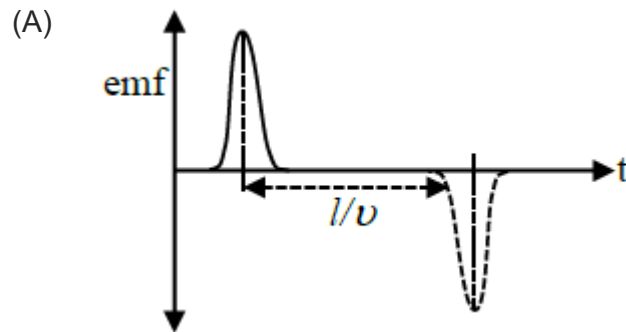
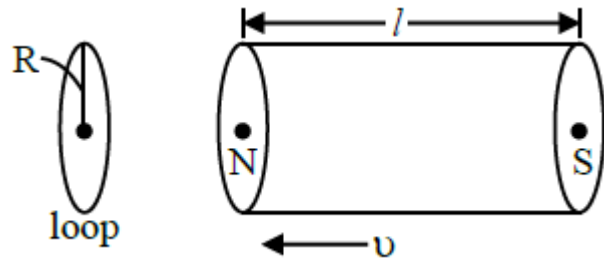
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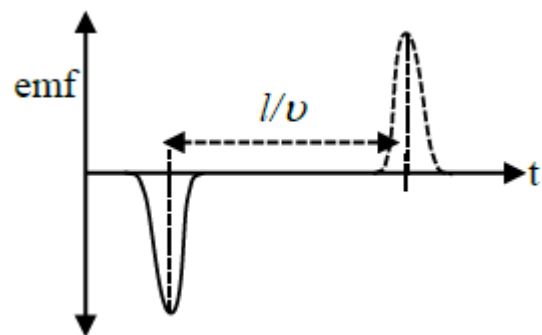
Solution:

- Increasing intensity means number of incident photons are increased.
- Kinetic energy of ejected electrons depend on the frequency of incident photons, not the intensity.

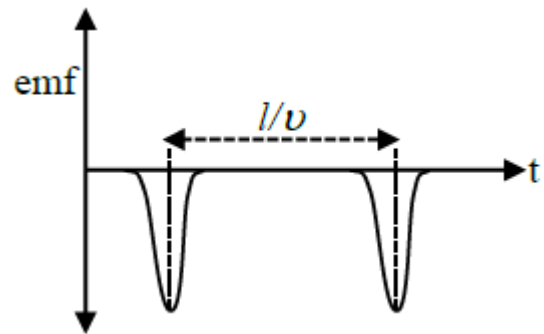
Q14: A bar magnet is passing through conducting loop of radius R with velocity v . The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve :



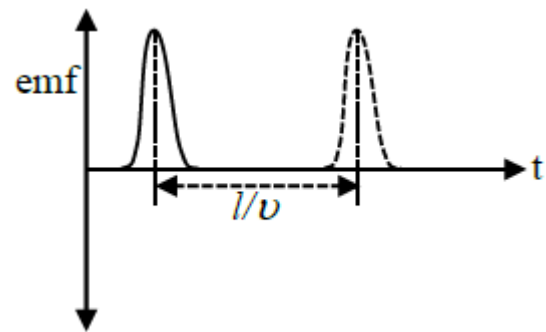
(B)



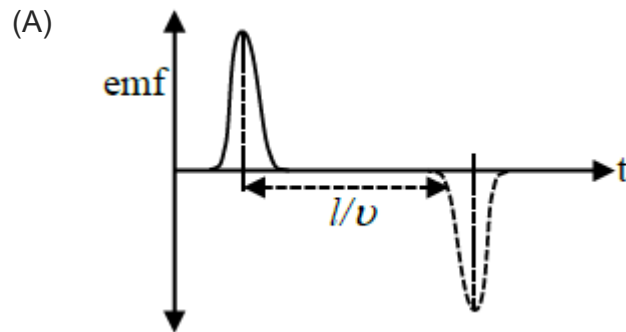
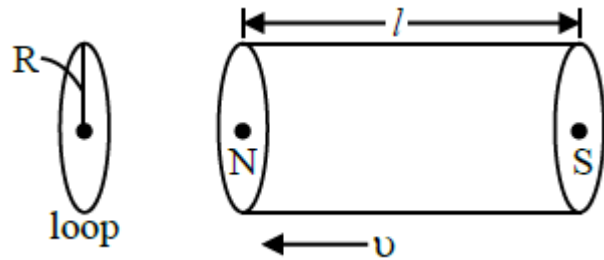
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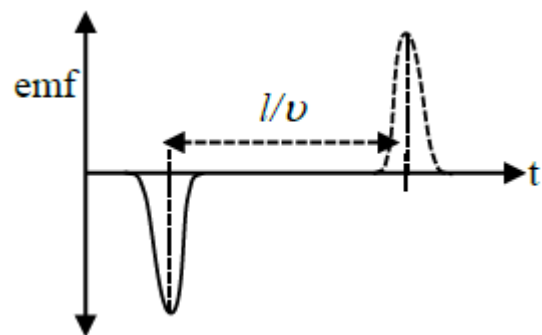
(D)



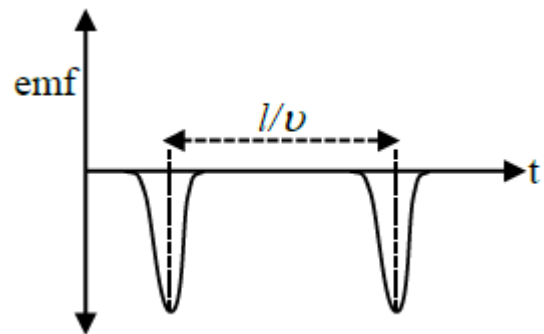
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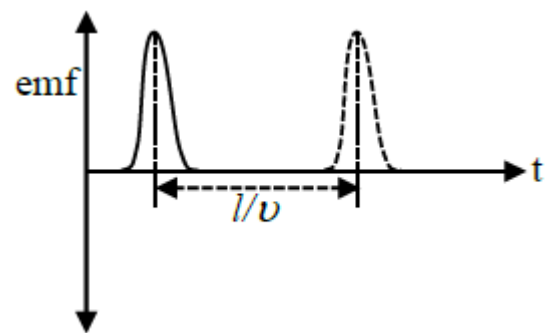
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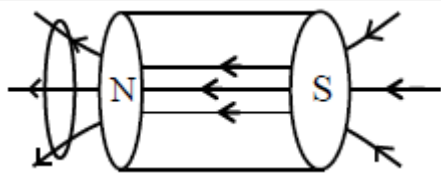
(C)



(D)



Solution:



- When a magnet passes through the central region of solenoid, no current/ Emf is induced in loop.
- While entering flux increases so negative induced emf
- While leaving flux decreases so positive induced emf.

Q15: Two ions of masses 4 amu and 16 amu have charges $+2e$ and $+3e$ respectively. These ions pass through the region of constant perpendicular magnetic field. The kinetic energy of both ions is same. Then ;

- (A) lighter ion will be deflected less than heavier ion
- (B) lighter ion will be deflected more than heavier ion
- (C) both ions will be deflected equally
- (D) no ion will be deflected.

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Solution:

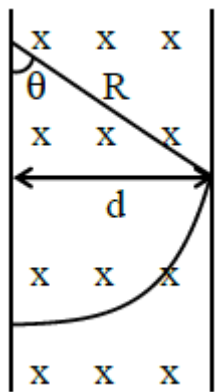
$$r = \frac{P}{qB} = \frac{\sqrt{2mk}}{qB}$$

Given they have same kinetic energy

$$r \propto \frac{\sqrt{m}}{q}$$

$$\frac{r_1}{r_2} = \frac{\sqrt{4}}{2} \times \frac{3}{\sqrt{16}} = \frac{3}{4}$$

$$r_2 = \frac{4r_1}{3} \quad (r_2 \text{ is for heavier ion and } r_1 \text{ is for lighter ion})$$



$$\sin \theta = \frac{d}{R}$$

$\theta \rightarrow$ Deflection

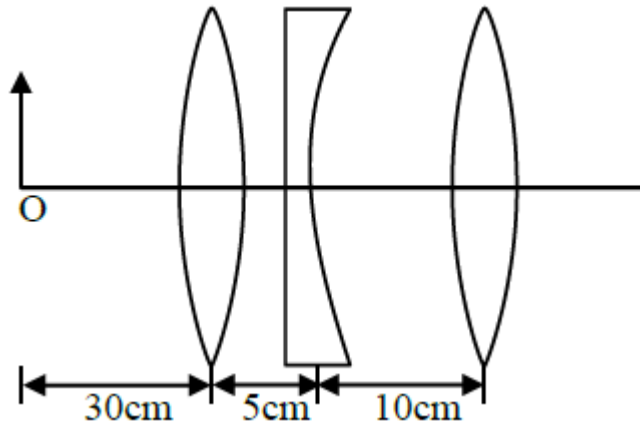
$$\theta \propto \frac{1}{R}$$

($R \rightarrow$ Radius of path)

$$\therefore R_2 > R_1 \Rightarrow \theta_2 < \theta_1$$

Q16: Find the distance of the image from object O, formed by the combination of lenses in the figure:

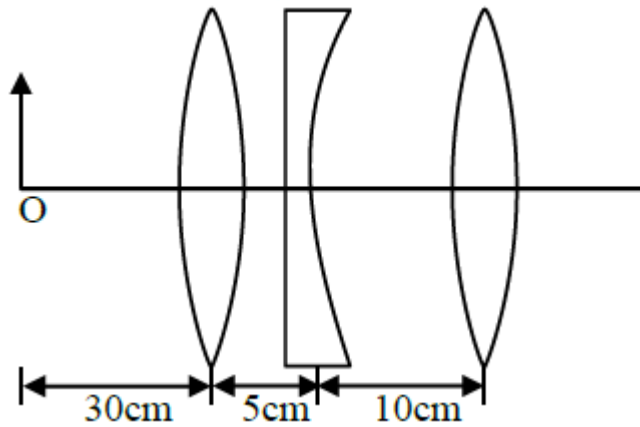
$$f = +10\text{cm} \quad f = -10\text{cm} \quad f = +30\text{cm}$$



- (A) 75 cm
- (B) 10 cm
- (C) 20 cm
- (D) infinity

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$$f = +10\text{cm} \quad f = -10\text{cm} \quad f = +30\text{cm}$$



- (A) 75 cm
- (B) 10 cm
- (C) 20 cm
- (D) infinity

Solution:

$$\frac{1}{V_1} + \frac{1}{30} = \frac{1}{10}$$

$$\frac{1}{V_1} = \frac{2}{30} \Rightarrow V_1 = 15 \text{ cm}$$

$$\frac{1}{V_2} - \frac{1}{10} = -\frac{1}{10}$$

$$\frac{1}{V_2} = 0$$

$$V_2 = \infty$$

$$V_3 = 30 \text{ cm}$$

$$OV_3 = 75 \text{ cm}$$

Q17: In Millikan's oil drop experiment, what is viscous force acting on an uncharged drop of radius $2.0 \times 10^{-5} \text{ m}$ and density $1.2 \times 10^3 \text{ kgm}^{-3}$? Take viscosity of liquid $= 1.8 \times 10^{-5} \text{ Nsm}^{-2}$. (Neglect buoyancy due to air).

- (A) $3.8 \times 10^{-11} \text{ N}$
- (B) $3.9 \times 10^{-10} \text{ N}$
- (C) $1.8 \times 10^{-10} \text{ N}$
- (D) $5.8 \times 10^{-10} \text{ N}$

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- (D) $5.8 \times 10^{-10} \text{ N}$

Solution:

$$\begin{aligned}\text{Viscous force} &= \text{Weight} \\ &= \rho \times \left(\frac{4}{3} \pi r^3 \right) g \\ &= 3.9 \times 10^{-10}\end{aligned}$$

Q18: Electric field in a plane electromagnetic wave is given by

$E = 50 \sin(500x - 10 \times 10^{10}t) V/m$. The velocity of electromagnetic wave in this medium is :

(Given C = speed of light in vacuum)

(A) $\frac{3}{2}C$

(B) C

(C) $\frac{2}{3}C$

(D) $\frac{C}{2}$

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(C) $\frac{2}{3}C$

(D) $\frac{C}{2}$

Solution:

$$V = \frac{\omega}{K} = \frac{10 \times 10^{10}}{500} = 2 \times 10^8$$
$$V = \frac{2C}{3}.$$

Q19: Five identical cells each of internal resistance 1Ω and emf 5V are connected in series and in parallel with an external resistance 'R'. For what value of 'R', current in series and parallel combination will remain the same ?

- (A) 1Ω
- (B) 25Ω
- (C) 5Ω
- (D) 10Ω

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Solution:

$$i_1 = \frac{25}{5 + R}$$

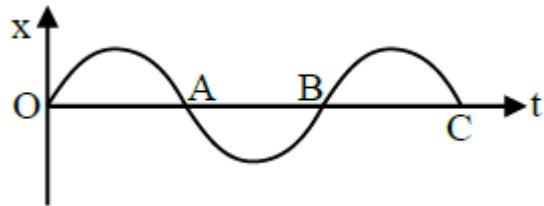
$$i_2 = \frac{5}{R + \frac{1}{5}}$$

$$i_1 = i_2 \Rightarrow 5 \left(R + \frac{1}{5} \right) = 5 + R$$

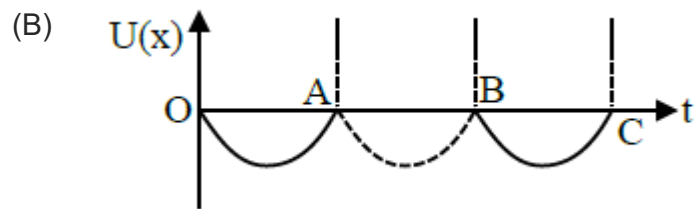
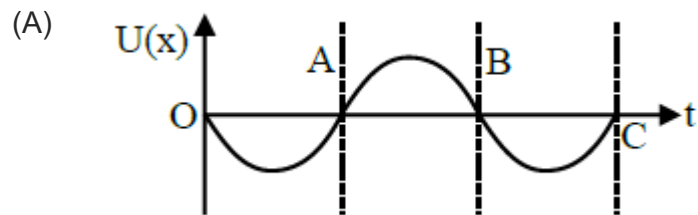
$$4R = 4$$

$$R = 1\Omega$$

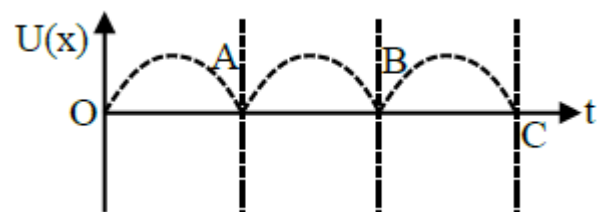
Q20: The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure.



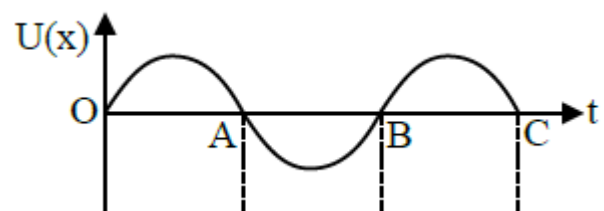
The potential energy $U(x)$ versus time (t) plot of the particle is correctly shown in figure :



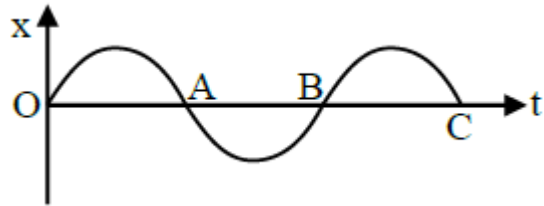
(C)



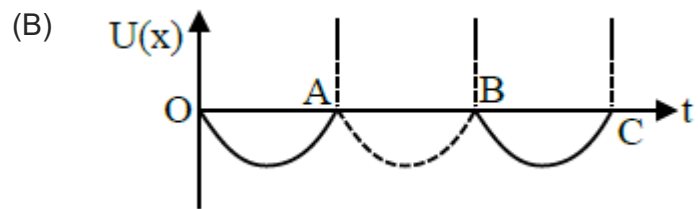
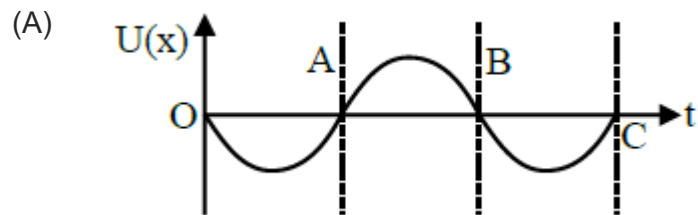
(D)



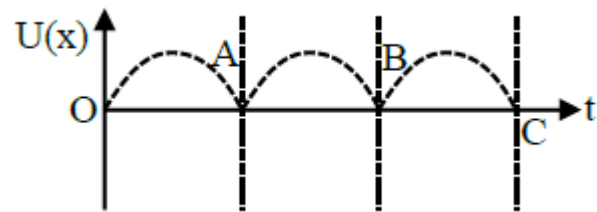
Q20: The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure.



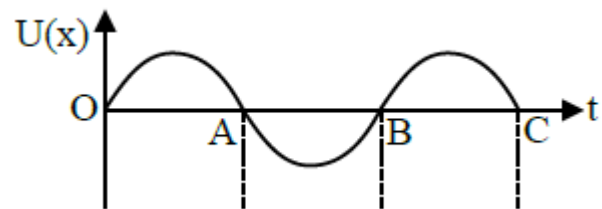
The potential energy $U(x)$ versus time (t) plot of the particle is correctly shown in figure :



(C)



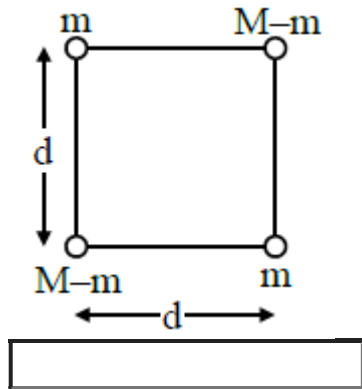
(D)



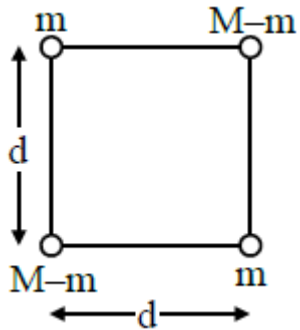
Solution:

Potential energy is maximum at maximum distance from mean.

Q21: A body of mass $(2M)$ splits into four masses $\{m, M - m, m, M - m\}$, which are rearranged to form a square as shown in the figure. The ratio of $\frac{M}{m}$ for which, the gravitational potential energy of the system becomes maximum is $x : 1$. The value of x is



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2.00

Solution:

$$u = \frac{Gm^2}{a} + \frac{G(M-m)^2}{a} + \frac{2Gm(M-m)}{a} + \frac{2Gm(M-m)}{\sqrt{2}a}$$

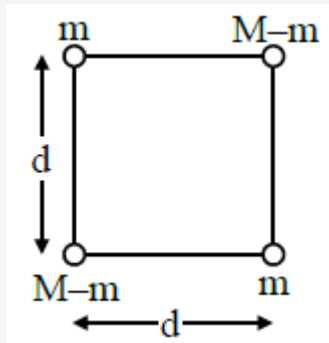
For maxima or minima of potential energy $(dU/dm) = 0$

$$dU/dm = (G/a)(2m - 2(M-m)) + (M-2m)^2 + \sqrt{2}(M-2m)$$

$$(4m - 4M + 2\sqrt{2}m) + (-2M + 2M + \sqrt{2}M) = 0$$

$$\sqrt{2}M = 2\sqrt{2}m$$

$$M/m = 2$$



Q22: The alternating current is given by

$$i = \left\{ \sqrt{42} \sin\left(\frac{2\pi}{T}t\right) + 10 \right\} A$$

The r.m.s. value of this current isA.

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The r.m.s. value of this current isA.

Solution:

$$\begin{aligned} f_{rms}^2 &= f_{1rms}^2 + f_{2rms}^2 \\ &= \left(\frac{\sqrt{42}}{\sqrt{2}} \right)^2 + 10^2 \\ f_{rms}^2 &= 121 \Rightarrow f_{rms} = 11 A \end{aligned}$$

Q23: A uniform conducting wire of length is $24a$, and resistance R is wound up as a current carrying coil in the shape of an equilateral triangle of side 'a' and then in the form of a square of side 'a'. The coil is connected to a voltage source V_0 . The ratio of magnetic moment of the coils in case of equilateral triangle to that for square is $1 : \sqrt{y}$ where y is

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3.00

Solution:

$$\text{In triangle shape } N_t = \frac{24a}{3a} = 8$$

$$\text{In square } N_s = \frac{24a}{4a} = 6$$

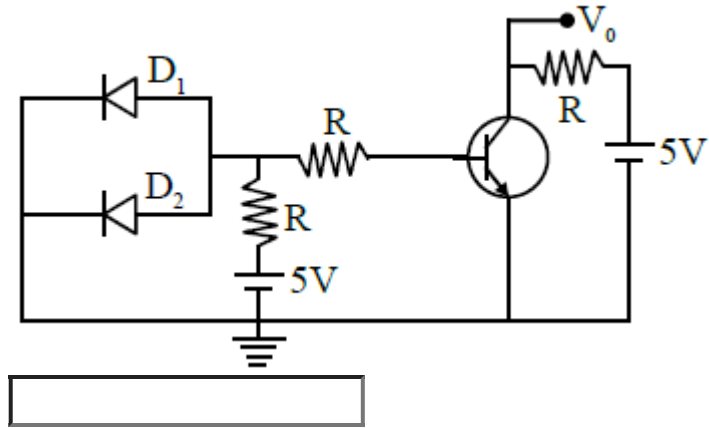
$$\frac{M_t}{M_s} = \frac{N_t I A_t}{N_s I A_s} \text{ [I will be same in both]}$$

$$= \frac{8 \times \frac{\sqrt{3}}{4} \times a^2}{6 \times a^2}$$

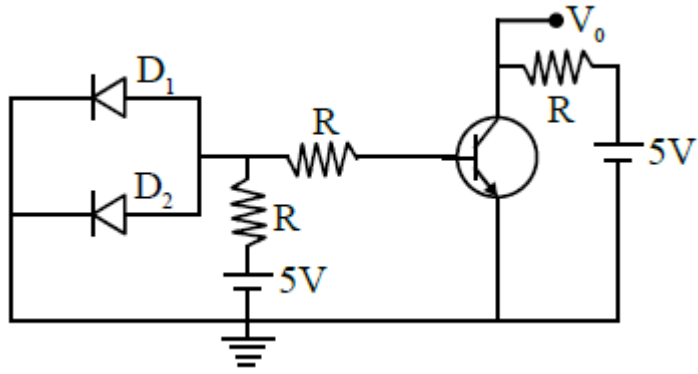
$$\frac{M_t}{M_s} = \frac{1}{\sqrt{3}}$$

$$\boxed{y = 3}$$

Q24: A circuit is arranged as shown in figure. The output voltage V_0 is equal toV.



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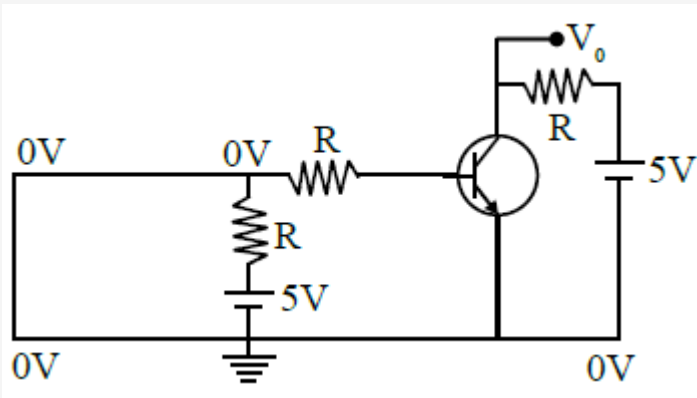


5.00

Solution:

As diodes D_1 and D_2 are in forward bias, so they acted as negligible resistances

$\Rightarrow$ Input voltage become zero



$\Rightarrow$ Input current is zero

$\Rightarrow$ Output current is zero

$\Rightarrow V_0 = 5 \text{ volt}$

Q25: First, a set of n equal resistors of $10\ \Omega$ each are connected in series to a battery of emf 20V and internal resistance $10\ \Omega$. A current I is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of n is

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20.00

Solution:

In series

$$R_{eq} = nR = 10n$$

$$i_s = \frac{20}{10 + 10n} = \frac{2}{1 + n}$$

in parallel

$$R_{eq} = \frac{10}{n}$$

$$i_p = \frac{20}{\frac{10}{n} + 10} = \frac{2n}{1 + n}$$

$$\frac{i_p}{i_s} = 20$$

$$\frac{\left(\frac{2n}{1+n}\right)}{\left(\frac{2}{1+n}\right)} = 20$$

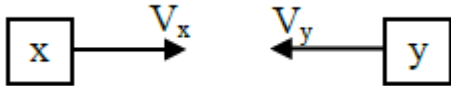
$$n = 20$$

Q26: Two cars X and Y are approaching each other with velocities 36 km/h and 72 km/h respectively. The frequency of a whistle sound as emitted by a passenger in car X, heard by the passenger in car Y is 1320 Hz. If the velocity of sound in air is 340 m/s, the actual frequency of the whistle sound produced is Hz.

Q26: Two cars X and Y are approaching each other with velocities 36 km/h and 72 km/h respectively. The frequency of a whistle sound as emitted by a passenger in car X, heard by the passenger in car Y is 1320 Hz. If the velocity of sound in air is 340 m/s, the actual frequency of the whistle sound produced is Hz.

1210.00

Solution:



$$V_x = 36 \text{ km/hr} = 10 \text{ m/s}$$

$$V_y = 72 \text{ km/hr} = 20 \text{ m/s}$$

by doppler's effect

$$F' = F_0 \left(\frac{V \pm V_0}{V \pm V_s} \right)$$

$$1320 = F_0 \left(\frac{340 + 20}{340 - 10} \right) \Rightarrow F_0 = 1210 \text{ Hz}$$

Q27: If the velocity of a body related to displacement x is given by $v = \sqrt{5000 + 24x} \text{ m/s}$, then the acceleration of the body is m/s^2 .

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12.00

Solution:

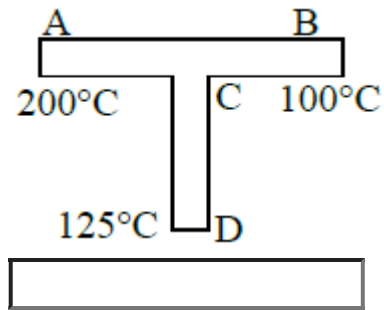
$$V = \sqrt{5000 + 24x}$$
$$\frac{dV}{dx} = \frac{1}{2\sqrt{5000 + 24x}} \times 24 = \frac{12}{\sqrt{5000 + 24x}}$$

$$\text{now } a = V \frac{dV}{dx}$$

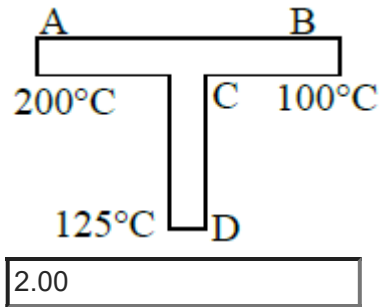
$$= \sqrt{5000 + 24x} \times \frac{12}{\sqrt{5000 + 24x}}$$

$$a = 12 \text{ m/s}^2$$

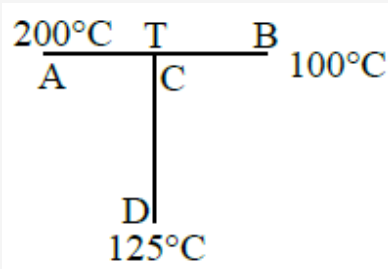
Q28: A rod CD of thermal resistance 10.0 KW^{-1} is joined at the middle of an identical rod AB as shown in figure, The end A, B and D are maintained at 200°C , 100°C and 125°C respectively. The heat current in CD is P watt. The value of P is



Q28: A rod CD of thermal resistance 10.0 KW^{-1} is joined at the middle of an identical rod AB as shown in figure, The end A, B and D are maintained at 200°C , 100°C and 125°C respectively. The heat current in CD is P watt. The value of P is



Solution:



Rods are identical so

$$R_{AB} = R_{CD} = 10 \text{ Kw}^{-1}$$

C is mid-point of AB, so

$$R_{AC} = R_{CB} = 5 \text{ Kw}^{-1}$$

at point C

$$\frac{200-T}{5} = \frac{T-125}{10} + \frac{T-100}{5}$$

$$2(100 - T) = T - 125 + 2(T - 100)$$

$$400 - 2T = T - 125 + 2T - 200$$

$$T = \frac{725}{5} = 145^\circ \text{C}$$

$$I_h = \frac{145-125}{10} w = \frac{20}{10} w$$

$$\boxed{I_h = 2w}$$

Q29: Two persons A and B perform same amount of work in moving a body through a certain distance d with application of forces acting at angle 45° and 60° with the direction of displacement respectively. The ratio of force applied by person A to the force applied by person B is $\frac{1}{\sqrt{x}}$. The value of x is

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2.00

Solution:

Given $W_A = W_B$

$$F_A d \cos 45^\circ = F_B d \cos 60^\circ$$

$$F_A \times \frac{1}{\sqrt{2}} = F_B \times \frac{1}{2}$$

$$\frac{F_A}{F_B} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$x = 2$$

Q30: A transmitting antenna has a height of 320 m and that of receiving antenna is 2000 m. The maximum distance between them for satisfactory communication in line of sight mode is 'd'. The value of 'd' is km.

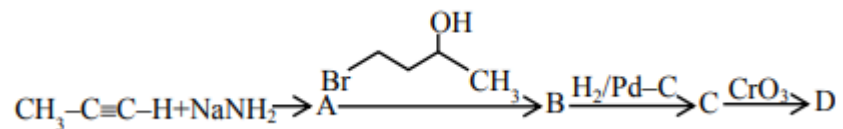
Q30: A transmitting antenna has a height of 320 m and that of receiving antenna is 2000 m. The maximum distance between them for satisfactory communication in line of sight mode is 'd'. The value of 'd' is km.

Solution:

$$d_m = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$
$$d_m = \left(\sqrt{2 \times 6400 \times 10^3 \times 320} + \sqrt{2 \times 6400 \times 10^3 \times 2000} \right) m$$

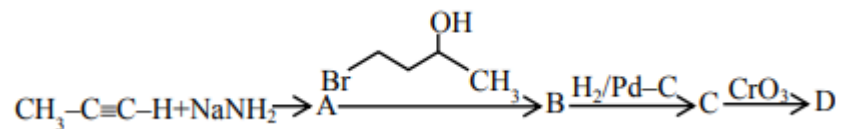
$$d_m = 224 \text{ km}$$

Q31: In the following sequence of reactions, the final product D is:



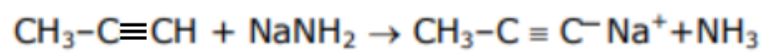
- (A) $\text{H}_3\text{C-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-C(=O)-H}$
- (B) $\text{CH}_3\text{-CH=CH-CH}_2\text{-CH}_2\text{-CH}_2\text{-COOH}$
- (C) $\text{H}_3\text{C-CH=CH-CH(OH)-CH}_2\text{-CH}_2\text{-CH}_3$
- (D) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-C(=O)-CH}_3$

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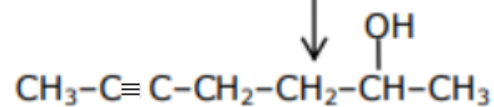
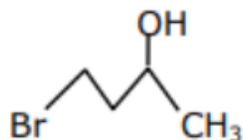


- (A) $\text{H}_3\text{C-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-C(=O)-H}$
- (B) $\text{CH}_3\text{-CH=CH-CH}_2\text{-CH}_2\text{-CH}_2\text{-COOH}$
- (C) $\text{H}_3\text{C-CH=CH-CH(OH)-CH}_2\text{-CH}_2\text{-CH}_3$
- (D) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-C(=O)-CH}_3$

Solution:

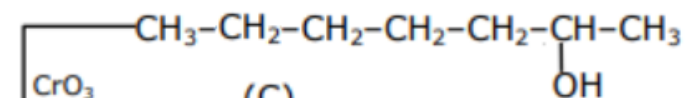


(A)



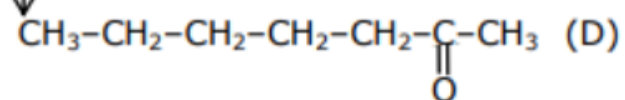
(B)

H₂/Pd-C



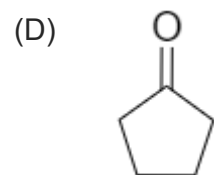
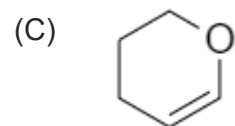
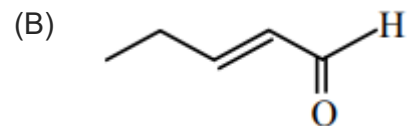
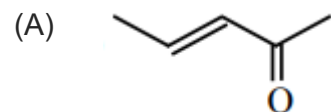
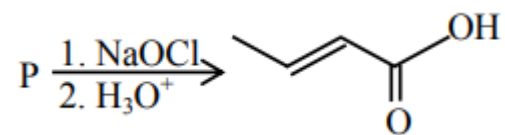
(C)

CrO₃

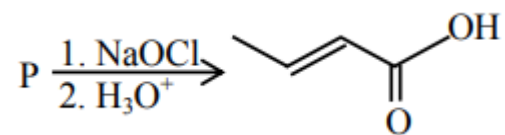


(D)

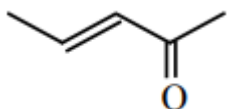
Q32: The structure of the starting compound P used in the reaction given below is



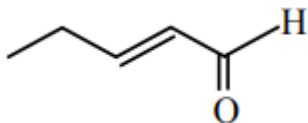
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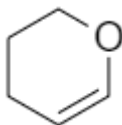
(A)



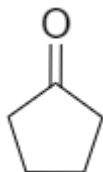
(B)



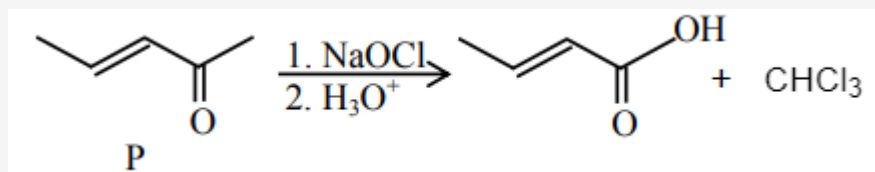
(C)



(D)



Solution:



NaOCl is used in haloform reaction as reagent.

Q33: Match **List-I** with **List-II**:

List – I(Species)	List – II(Number of lone pairs of electrons on the central atom)
(p) XeF_2	(i) 0
(q) XeO_2F_2	(ii) 1
(r) XeO_3F_2	(iii) 2
(s) XeF_4	(iv) 3

Choose the **most appropriate** answer from the options given below :

- (A) (p)–(iv), (q)–(i), (r)–(ii), (s)–(iii)
- (B) (p)–(iii), (q)–(iv), (r)–(ii), (s)–(i)
- (C) (p)–(iii), (q)–(ii), (r)–(iv), (s)–(i)
- (D) (p)–(iv), (q)–(ii), (r)–(i), (s)–(iii)

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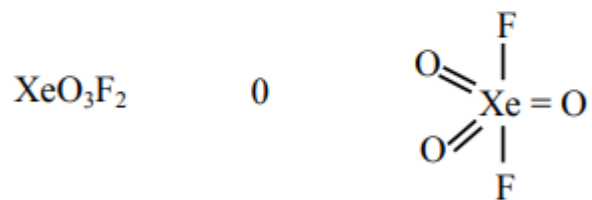
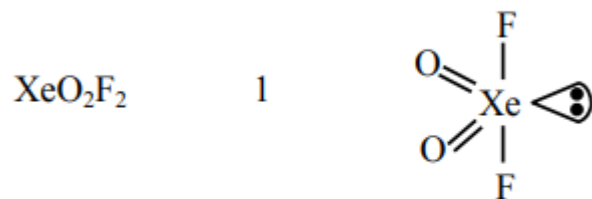
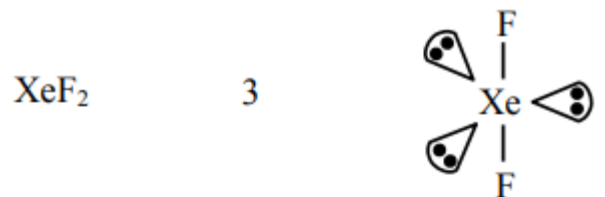
(B) (p)–(iii), (q)–(iv), (r)–(ii), (s)–(i)

(C) (p)–(iii), (q)–(ii), (r)–(iv), (s)–(i)

(D) (p)–(iv), (q)–(ii), (r)–(i), (s)–(iii)

Solution:

The species, the number of lone pairs on the central atom and the structures are as given below –



Q34: In which one of the following molecules strongest back donation of an electron pair from halide to boron is expected?

(A) BCl_3

(B) BF_3

(C) BBr_3

(D) BI_3

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Solution:

Compound	Type of back bonding
BF_3	$2p\pi-2p\pi$
BCl_3	$2p\pi-3p\pi$
BBr_3	$2p\pi-4p\pi$
BI_3	$2p\pi-5p\pi$

As the size of p-orbital increases, the strength of bond decreases. Therefore, back bonding strength is as follows - $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BI}_3$.

Q35: Deuterium resembles hydrogen in properties but:

- (A) reacts slower than hydrogen
- (B) reacts vigorously than hydrogen
- (C) reacts just as hydrogen
- (D) emits β^+ particles

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Solution:

The bond dissociation energy of D_2 is greater than H_2 and therefore D_2 reacts slower than H_2 .

Q36: Which refining process is generally used in the purification of low melting metals?

- (A) Chromatographic method
- (B) Liquation
- (C) Electrolysis
- (D) Zone refining

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- (D) Zone refining

Solution:

Liquation method is used to purify those impure metals which has lower melting point than the melting point of impurities associated. $\therefore$ This method is used for metal having low melting point.

Q37: Match items of **List-I** with those of **List-II**:

List – I(Property)	List – II(Example)
(p) Diamagnetism	(i) MnO
(q) Ferrimagnetism	(ii) O ₂
(r) Paramagnetism	(iii) NaCl
(s) Antiferromagnetism	(iv) Fe ₃ O ₄

Choose the **most appropriate** answer from the options given below:

- (A) (p)–(ii), (q)–(i), (r)–(iii), (s)–(iv)
- (B) (p)–(i), (q)–(iii), (r)–(iv), (s)–(ii)
- (C) (p)–(iii), (q)–(iv), (r)–(ii), (s)–(i)
- (D) (p)–(iv), (q)–(ii), (r)–(i), (s)–(iii)

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(s) Antiferromagnetism	(iv) Fe ₃ O ₄

Choose the **most appropriate** answer from the options given below:

(A) (p)–(ii), (q)–(i), (r)–(iii), (s)–(iv)

(B) (p)–(i), (q)–(iii), (r)–(iv), (s)–(ii)

(C) (p)–(iii), (q)–(iv), (r)–(ii), (s)–(i)

(D) (p)–(iv), (q)–(ii), (r)–(i), (s)–(iii)

Solution:

According to NCERT data-

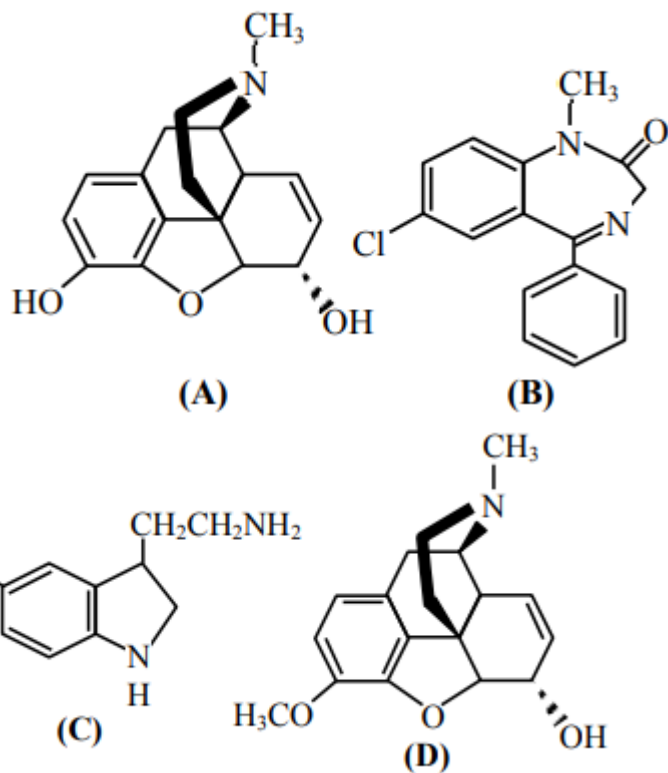
Diamagnetism is shown by NaCl.

Ferrimagnetism is shown by Fe₃O₄.

Paramagnetism is shown by O₂.

Antiferromagnetism is shown by MnO

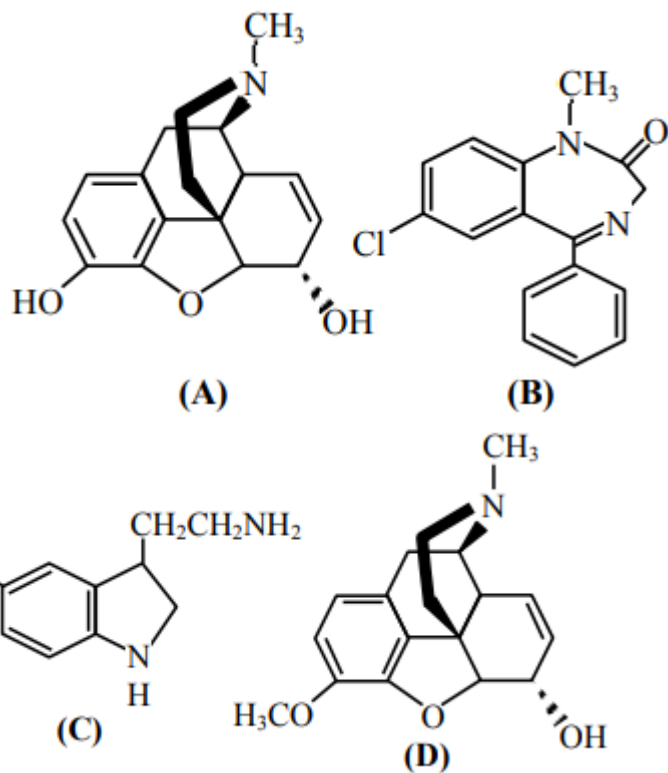
Q38:



The correct statement about (A), (B), (C) and (D) is

- (A) (A), (B) and (C) are narcotic analgesics
- (B) (B), (C) and (D) are tranquillizers
- (C) (A) and (D) are tranquillizers
- (D) (B) and (C) are tranquillizers

Q38:



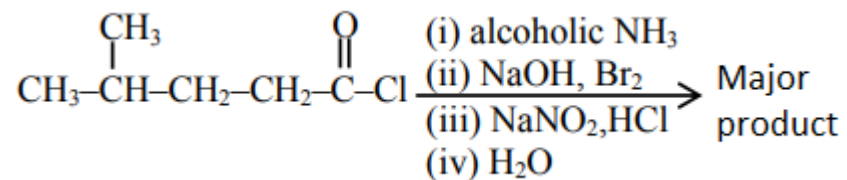
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Solution:

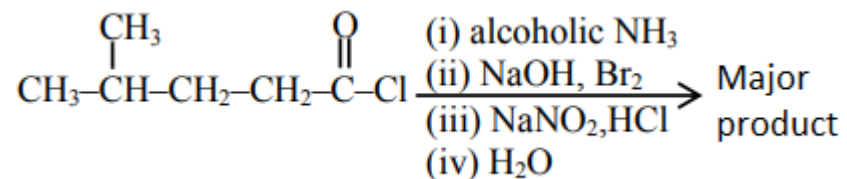
B is valium and C is serotonin are tranquilizers. While A and D are Morphine and Codeine are narcotic analgesics.

Q39: The major product of the following reaction is



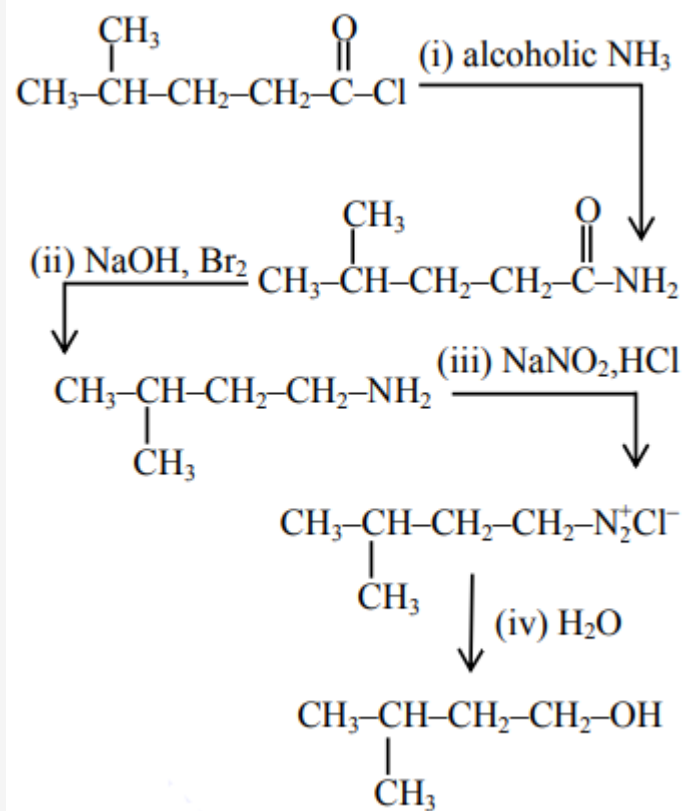
- (A) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\overset{\text{Br}}{\underset{|}{\text{CH}}}-\text{CH}_2\text{OH}$
- (B) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2-\text{CH}_2\text{OH}$
- (C) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2\text{OH}$
- (D) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2-\text{Cl}$

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- (A) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\overset{\text{Br}}{\underset{|}{\text{CH}}}-\text{CH}_2\text{OH}$
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- (C) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2\text{OH}$
- (D) $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2-\text{Cl}$

Solution:



Q40: Which of the following is not a correct statement for primary aliphatic amines?

- (A) The intermolecular association in primary amines is less than the intermolecular association in secondary amines.
- (B) Primary amines on treating with nitrous acid solution form corresponding alcohols except methyl amine.
- (C) Primary amines are less basic than the secondary amines.
- (D) Primary amines can be prepared by the Gabriel phthalimide synthesis.

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Solution:

The intermolecular association is more prominent in case of primary amines as compared to secondary, due to the availability of two hydrogen atom.

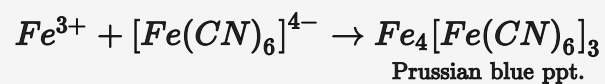
Q41: Acidic ferric chloride solution on treatment with excess of potassium ferrocyanide gives a Prussian blue coloured ppt. It is:

- (A) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- (B) $\text{K}_3\text{Fe}[\text{Fe}(\text{CN})_6]_2$
- (C) $\text{HFe}[\text{Fe}(\text{CN})_6]$
- (D) $\text{KFe}[\text{Fe}(\text{CN})_6]$

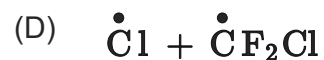
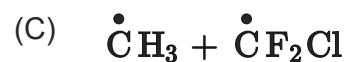
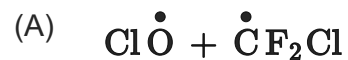
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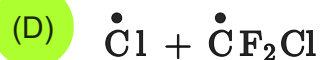
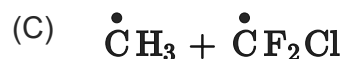
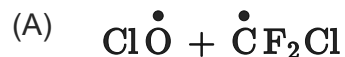
Solution:



Q42: The gas 'A' is having very low reactivity reaches to stratosphere. It is non-toxic and non-flammable but dissociated by UV—radiations in stratosphere. The intermediates formed initially from the gas 'A' are :



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Solution:

In stratosphere CFCs get broken down by powerful UV radiations releasing $\dot{\text{C}}\text{l}$

$$\text{CF}_2\text{Cl}_2(\text{g}) \xrightarrow{\text{U.V}} \text{Cl}^\bullet(\text{g}) + \bullet\text{CF}_2\text{Cl}(\text{g})$$

Q43: The number of water molecules in gypsum, dead burnt plaster and plaster of Paris, respectively are :

- (A) 2, 0 and 1
- (B) 0.5, 0 and 2
- (C) 5, 0 and 0.5
- (D) 2, 0 and 0.5

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- (A) 2, 0 and 1
- (B) 0.5, 0 and 2
- (C) 5, 0 and 0.5
- (D) 2, 0 and 0.5

Solution:

Gypsum - $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Dead burnt plaster - CaSO_4

Plaster of Paris - $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$

Q44: The nature of oxides V_2O_3 and CrO is indexed as 'X' and 'Y' type respectively. The correct set of X and Y is

- (A) X = basic, Y = amphoteric
- (B) X = amphoteric, Y = basic
- (C) X = acidic, Y = acidic
- (D) X = basic, Y = basic

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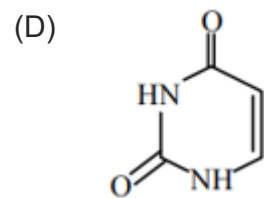
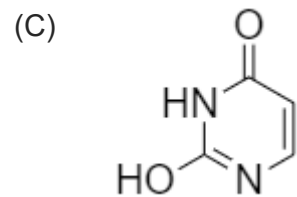
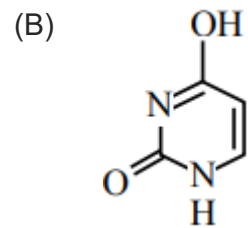
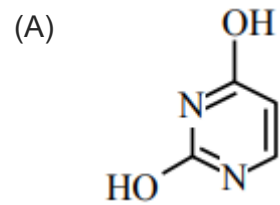
Solution:

In case of transition metal oxides, the oxides with metals in lower oxidation states are basic in nature.

V_2O_3 - basic

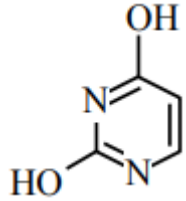
CrO – basic

Q45: Out of following isomeric forms of uracil, which one is present in RNA?

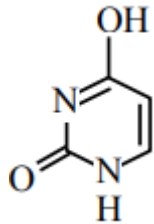


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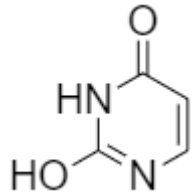
(A)



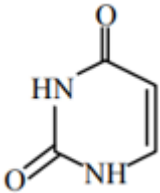
(B)



(C)

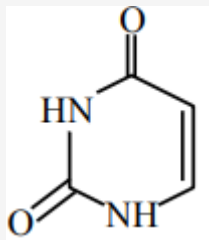


(D)



Solution:

Isomeric form of uracil present in RNA



Q46: Given below are two statements: one is labelled as :**Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Synthesis of ethyl phenyl ether may be achieved by Williamson synthesis.

Reason (R): Reaction of bromobenzene with sodium ethoxide yields ethyl phenyl ether.

In the light of the above statements, choose the **most appropriate** answer from the options given below

- (A) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
- (B) **(A)** is correct but **(R)** is not correct
- (C) **(A)** is not correct but **(R)** is correct
- (D) Both **(A)** and **(R)** are correct but **(R)** is NOT the correct explanation of **(A)**

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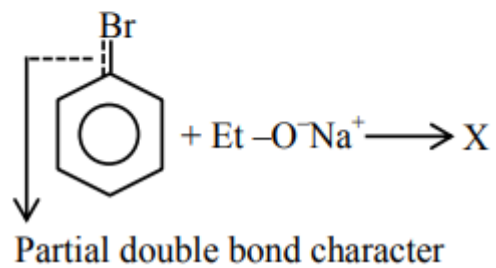
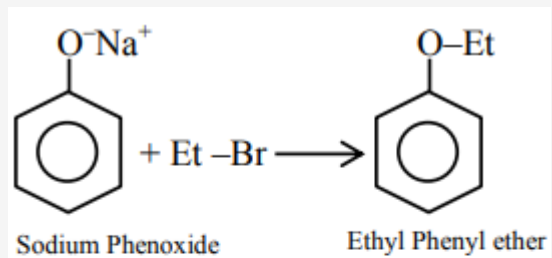
Assertion (A): Synthesis of ethyl phenyl ether may be achieved by Williamson synthesis.

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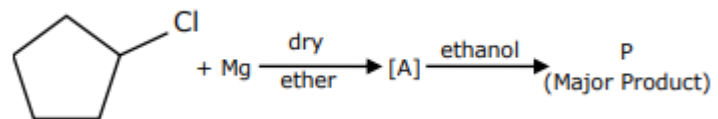
In the light of the above statements, choose the **most appropriate** answer from the options given below

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Solution:



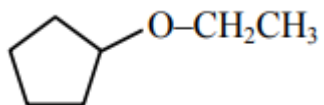
Q47: In the following sequence of reactions the P is :



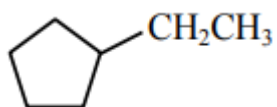
(A)



(B)



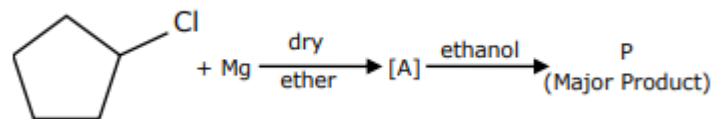
(C)



(D)



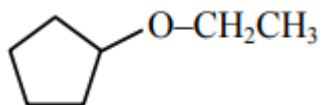
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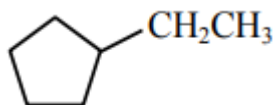
(A)



(B)



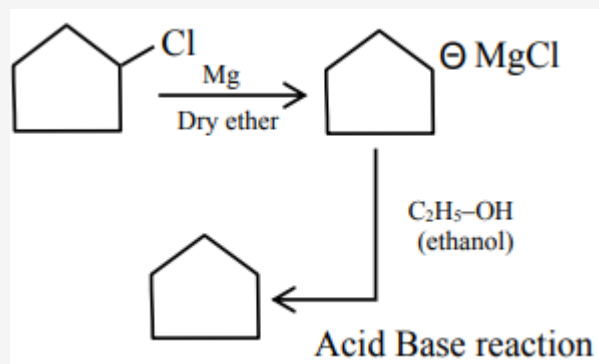
(C)



(D)



Solution:



Q48: The unit of the van der Waals gas equation parameter 'a' in

$$\left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT \text{ is:}$$

- (A) kg m s^{-2}
- (B) $\text{dm}^3 \text{ mol}^{-1}$
- (C) kg m s^{-1}
- (D) $\text{atm dm}^6 \text{ mol}^{-2}$

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- (C) kg m s^{-1}
- (D) $\text{atm dm}^6 \text{ mol}^{-2}$

Solution:

The unit of $\frac{an^2}{V^2}$ is same as the unit of pressure.

$$\frac{an^2}{V^2} = atm \Rightarrow a = atm \times \frac{dm^6}{mol^2}$$

Q49: In polythionic acid, $\text{H}_2\text{S}_x\text{O}_6$ ($x = 3$ to 5) the oxidation state(s) of sulphur is/are:

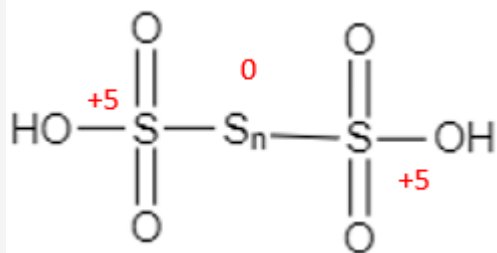
- (A) + 5 only
- (B) + 6 only
- (C) + 3 and + 5 only
- (D) 0 and + 5 only

Q49: In polythionic acid, $\text{H}_2\text{S}_x\text{O}_6$ ($x = 3$ to 5) the oxidation state(s) of sulphur is/are:

- (A) + 5 only
- (B) + 6 only
- (C) + 3 and + 5 only
- (D) 0 and + 5 only

Solution:

The general structure of polythionic acids is –



Here $n = 1$ to 3 . Thus, possible oxidation states of sulphur are 0 and +5 only.

Q50: Tyndall effect is more effectively shown by:

- (A) true solution
- (B) lyophilic colloid
- (C) lyophobic colloid
- (D) suspension

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- (A) true solution
- (B) lyophilic colloid
- ☒ (C) lyophobic colloid
- (D) suspension

Solution:

Tyndall effect occurs due to scattering of light by the colloidal particles. In the case of lyophilic colloids there is not much difference in the refractive index of the dispersed phase and dispersion medium, so scattering is very less and Tyndall effect is almost unobservable in such cases.

Q51: In Carius method for estimation of halogens, 0.2 g of an organic compound gave 0.188 g of AgBr. The percentage of bromine in the compound is _____. (Nearest integer)

[Atomic mass : Ag = 108, Br = 80]

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[Atomic mass : Ag = 108, Br = 80]

40.00

Solution:

$$n_{\text{AgBr}} = \frac{0.188\text{g}}{188\text{g/mol}} = 10^{-3}\text{mol}$$

$$\Rightarrow n_{\text{Br}} = n_{\text{AgBr}} = 0.01\text{ mol}$$

$$\Rightarrow \text{mass}_{\text{Br}} = (0.01 \times 80)\text{gm} = 0.08\text{ gm}$$

$$\Rightarrow \text{mass \%} = \frac{0.08 \times 100}{0.2} = 40\%$$

Q52: The reaction that occurs in a breath analyzer, a device used to determine the alcohol level in a person's blood stream is:



If the rate of appearance of $\text{Cr}_2(\text{SO}_4)_3$ is $2.67 \text{ mol min}^{-1}$ at a particular time, the rate of disappearance of $\text{C}_2\text{H}_6\text{O}$ at the same time is _____ mol min^{-1} . (Nearest integer)

Q52: The reaction that occurs in a breath analyzer, a device used to determine the alcohol level in a person's blood stream is:



If the rate of appearance of $\text{Cr}_2(\text{SO}_4)_3$ is $2.67 \text{ mol min}^{-1}$ at a particular time, the rate of disappearance of $\text{C}_2\text{H}_6\text{O}$ at the same time is _____ mol min^{-1} . (Nearest integer)

4.00

Solution:

$$\left(\frac{\text{Rate of disappearance of } \text{C}_2\text{H}_6\text{O}}{3} \right) = \left(\frac{\text{Rate of appearance of } \text{Cr}_2(\text{SO}_4)_3}{2} \right)$$

$$\Rightarrow \left(\frac{2.67 \text{ mol / min} \times 3}{2} \right) = \text{rate of disappearance of } \text{C}_2\text{H}_6\text{O}$$

$$\Rightarrow \text{Rate of disappearance of } \text{C}_2\text{H}_6\text{O} = 4.005 \text{ mol/min}$$

Q53: The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is equal to $\frac{h^2}{xma_0^2}$. The value of 10x is _____. (Nearest integer) (a_0 is radius of Bohr's orbit)

[Given: $\pi = 3.14$]

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[Given: $\pi = 3.14$]

3155.00

Solution:

$$mvr = \frac{nh}{2\pi} \Rightarrow v^2 = \left(\frac{nh}{2\pi mr} \right)^2$$

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{n^2h^2}{8\pi^2mr^2} = \frac{4h^2}{8\pi^2m(4a_0)^2} = \left(\frac{4}{8\pi^2 \times 16} \right) \frac{h^2}{ma_0^2}$$

$$\Rightarrow x = 315.507$$

$$\Rightarrow 10x = 3155 \text{ (nearest integer)}$$

Q54: 1 kg of 0.75 molal aqueous solution of sucrose can be cooled up to -4°C before freezing. The amount of ice (in g) that will be separated out is _____. (Nearest integer)
[Given : $K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$]

Q54: 1 kg of 0.75 molal aqueous solution of sucrose can be cooled up to -4°C before freezing. The amount of ice (in g) that will be separated out is _____. (Nearest integer)
 [Given : $K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$]

518.00

Solution:

Let mass of water initially present = x gm

Mass of sucrose = $(1000 - x)$ gm

Moles of sucrose = $\left(\frac{1000-x}{342}\right)$

$$\Rightarrow 0.75 = \frac{\left(\frac{1000-x}{342}\right)}{\left(\frac{x}{1000}\right)} \Rightarrow \frac{x}{1000} = \frac{1000-x}{342 \times 0.75}$$

$$\Rightarrow 256.6x = 10^6 - 1000x$$

$$\Rightarrow x = 795.86 \text{ gm}$$

$$\Rightarrow \text{moles of sucrose} = 0.5969$$

New mass of H_2O = a kg

$$\Rightarrow 4 = \frac{0.5969}{a} \times 1.86 \Rightarrow a = 0.2775 \text{ kg}$$

$$\Rightarrow \text{ice separated} = (795.86 - 277.5) = 518.3 \text{ gm}$$

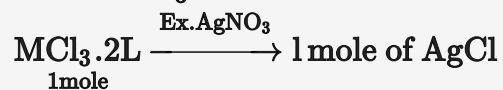
Q55: 1 mol of an octahedral metal complex with formula $\text{MCl}_3 \cdot 2\text{L}$ on reaction with excess of AgNO_3 gives 1 mol of AgCl . The denticity of Ligand L is _____. (Integer answer)

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2.00

Solution:

$\text{MCl}_3 \cdot 2\text{L}$ octahedral



Its means that one Cl^- ion present in ionization sphere.

Thus, formula = $[\text{MCl}_2\text{L}_2]\text{Cl}$

For octahedral complex coordination no. is 6. Therefore, L act as bidentate ligand.

Q56: The number of moles of CuO, that will be utilized in Dumas method for estimation nitrogen in a sample of 57.5 g of N, N-dimethylaminopentane is _____ $\times 10^{-2}$. (Nearest integer):

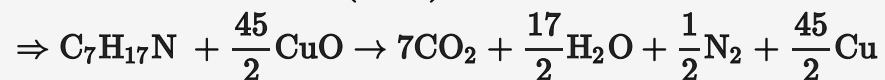
Q56: The number of moles of CuO, that will be utilized in Dumas method for estimation nitrogen in a sample of 57.5 g of N, N-dimethylaminopentane is _____ $\times 10^{-2}$. (Nearest integer):

1125.00

Solution:

Moles of N in N,N - dimethylaminopentane

$$= \left(\frac{57.5}{115} \right) = 0.5 \text{ mol}$$



$$\frac{n_{\text{CuO reacted}}}{\left(\frac{45}{2}\right)} = \frac{n_{\text{C}_7\text{H}_{17}\text{N reacted}}}{1}$$

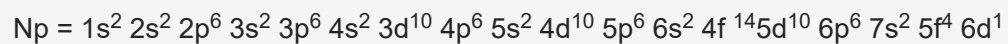
$$\Rightarrow n_{\text{CuO reacted}} = \left(\frac{45}{2} \right) \times 0.5 = 11.25$$

Q57: The number of f electrons in the ground state electronic configuration of Np ($Z = 93$) is _____. (Nearest integer)

Q57: The number of f electrons in the ground state electronic configuration of Np (Z = 93) is _____. (Nearest integer)

18.00

Solution:



Total no. of 'f' electrons = 14 + 4 = 18

Q58: 200 mL of 0.2 M HCl is mixed with 300 mL of 0.1 M NaOH. The molar heat of neutralization of this reaction is -57.1 kJ. The increase in temperature in $^{\circ}\text{C}$ of the system on mixing is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)

[Given: Specific heat of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ Density of water = 1.00 g cm^{-3}]

(Assume no volume change on mixing) :

Q58: 200 mL of 0.2 M HCl is mixed with 300 mL of 0.1 M NaOH. The molar heat of neutralization of this reaction is -57.1 kJ. The increase in temperature in $^{\circ}\text{C}$ of the system on mixing is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)

[Given: Specific heat of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ Density of water = 1.00 g cm^{-3}]

(Assume no volume change on mixing) :

82.00

Solution:

Millimoles of HCl = $200 \times 0.2 = 40$

Millimoles of NaOH = $300 \times 0.1 = 30$

Heat released = $\left(\frac{30}{1000} \times 57.1 \times 1000\right) = 1713 \text{ J}$

Mass of solution = $500 \text{ ml} \times 1 \text{ gm/ml} = 500 \text{ gm}$

$$\Rightarrow \Delta T = \frac{q}{m \times C} = \frac{1713 \text{ J}}{500 \text{ g} \times 4.18 \frac{\text{J}}{\text{g} \cdot \text{K}}} = 0.8196 \text{ K}$$
$$= 81.96 \times 10^{-2} \text{ K}$$

Q59: The number of moles of NH_3 , that must be added to 2 L of 0.80 M AgNO_3 in order to reduce the concentration of Ag^+ ions to 5.0×10^{-8} M ($K_{\text{formation}}$ for $[\text{Ag}(\text{NH}_3)_2]^+ = 1.0 \times 10^8$) is _____ . (Nearest integer)

[Assume no volume change on adding NH_3]

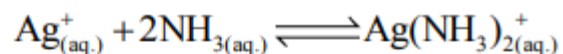
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[Assume no volume change on adding NH_3]

4.00

Solution:

Let moles added = a



$$t = 0 \quad 0.8 \quad \left(\frac{a}{2}\right)$$

$$t = \infty \quad 5 \times 10^{-8} \quad \left(\frac{a}{2} - 1.6\right) \quad 0.8$$

Writing the expression for equilibrium constant –

$$K_f = \frac{0.8}{(5 \times 10^{-8}) \left(\frac{a}{2} - 1.6\right)^2} = 10^8$$

$$\Rightarrow \frac{a}{2} - 1.6 = 0.4 \Rightarrow a = 4$$

Q60: When 10 mL of an aqueous solution of KMnO_4 was titrated in acidic medium, equal volume of 0.1 M of an aqueous solution of ferrous sulphate was required for complete discharge of colour. The strength of KMnO_4 in grams per litre is _____ $\times 10^{-2}$. (Nearest integer)

[Atomic mass of K = 39, Mn = 55, O = 16]

Q60: When 10 mL of an aqueous solution of KMnO_4 was titrated in acidic medium, equal volume of 0.1 M of an aqueous solution of ferrous sulphate was required for complete discharge of colour. The strength of KMnO_4 in grams per litre is _____ $\times 10^{-2}$. (Nearest integer)

[Atomic mass of K = 39, Mn = 55, O = 16]

316.00

Solution:

Let the molarity of KMnO_4 solution is x.

The n-factor for KMnO_4 in acidic medium is 5.

The n-factor for FeSO_4 is 1.

For complete reaction, the equivalents of the reacting substances must be equal.

$$n_1 M_1 V_1 = n_2 M_2 V_2$$

$$\Rightarrow 5 \times M_1 \times 10 = 1 \times 0.1 \times 10$$

$$\Rightarrow M_1 = \frac{0.1}{5} = \frac{1}{50}$$

$$\text{Now, the strength in gram per liter} = \frac{1}{50} \times 158 = 3.16 = 316 \times 10^{-2}$$

Q61: Let A be a fixed point $(0, 6)$ and B be a moving point $(2t, 0)$. Let M be the mid-point of AB and the perpendicular bisector of AB meets the y -axis at C . The locus of the mid-point P of MC is:

(A) $3x^2 + 2y - 6 = 0$

(B) $2x^2 + 3y - 9 = 0$

(C) $2x^2 - 3y + 9 = 0$

(D) $3x^2 - 2y - 6 = 0$

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(D) $3x^2 - 2y - 6 = 0$

Solution:

We have, $A(0, 6)$ and $B(2t, 0)$

The midpoint of AB is $M = (t, 3)$

Slope of $AB = \frac{-3}{t}$

Perpendicular bisector of AB is

$$(y - 3) = \frac{t}{3}(x - t)$$

$$\text{So, } C = \left(0, 3 - \frac{t^2}{3}\right)$$

Let P be (h, k)

$$\therefore h = \frac{t}{2} ; k = \left(3 - \frac{t^2}{6}\right)$$

$$\Rightarrow k = 3 - \frac{4h^2}{6}$$

$$\Rightarrow 2x^2 + 3y - 9 = 0$$

Q62: A tangent and a normal are drawn at the point $P(2, -4)$ on the parabola $y^2 = 8x$, which meet the directrix of the parabola at the points A and B respectively. If $Q(a, b)$ is a point such that $AQBP$ is a square, then $2a + b$ is equal to:

- (A) -16
- (B) -20
- (C) -18
- (D) -12

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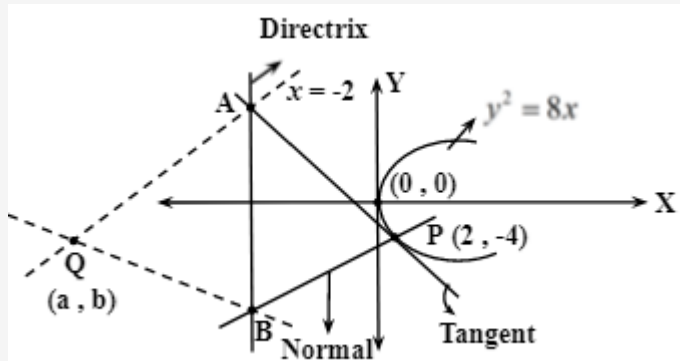
(A) -16

(B) -20

(C) -18

(D) -12

Solution:



Equation of tangent at $P(2, -4)$ is $-4y = 4(x + 2)$

$$\Rightarrow x + y + 2 = 0 \quad \dots (i)$$

Equation of normal is $y - (-4) = 1(x - 2)$

$$\Rightarrow y + 4 = x - 2$$

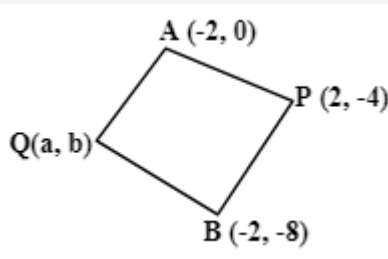
$$\Rightarrow y = x - 6 \quad \dots (ii)$$

Directrix is $x = -2$

$\therefore A = (-2, 0)$ and $B = (-2, -8)$

$P = (2, -4), Q(a, b)$

Given $AQBP$ is a square



Midpoint of PQ = Midpoint of AB

$$\left(\frac{a+2}{2}, \frac{b-4}{2} \right) = \left(\frac{-2-2}{2}, \frac{0-8}{2} \right)$$

$$\frac{a+2}{2} = -2 \text{ and } \frac{b-4}{2} = -4$$

$$a = -6 \text{ and } b - 4 = -8$$

$$b = -4$$

$$\text{Thus, } 2a + b = 2(-6) - 4 = -16$$

Q63: If α, β are the distinct roots of $x^2 + bx + c = 0$, then $\lim_{x \rightarrow \beta} \frac{e^{2(x^2+bx+c)} - 1 - 2(x^2+bx+c)}{(x-\beta)^2}$ is equal to:

- (A) $b^2 + 4c$
- (B) $2(b^2 - 4c)$
- (C) $b^2 - 4c$
- (D) $2(b^2 + 4c)$

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- (B) $2(b^2 - 4c)$
- (C) $b^2 - 4c$
- (D) $2(b^2 + 4c)$

Solution:

$$\begin{aligned}
 & \lim_{x \rightarrow \beta} \frac{e^{2(x^2+bx+c)} - 1 - 2(x^2+bx+c)}{(x-\beta)^2} \\
 & \Rightarrow \lim_{x \rightarrow \beta} \frac{\left(1 + \frac{2(x^2+bx+c)}{1!} + \frac{2^2(x^2+bx+c)^2}{2!} + \dots\right) - 1 - 2(x^2+bx+c)}{(x-\beta)^2} \\
 & \Rightarrow \lim_{x \rightarrow \beta} \frac{2(x^2+bx+c)^2}{(x-\beta)^2} \\
 & \Rightarrow \lim_{x \rightarrow \beta} \frac{2(x-\alpha)^2(x-\beta)^2}{(x-\beta)^2} \\
 & \Rightarrow 2(\beta - \alpha)^2 = 2(b^2 - 4c)
 \end{aligned}$$

Q64: Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx} = 2(y + 2 \sin x - 5)x - 2 \cos x$ such that $y(0) = 7$. then $y(\pi)$ is equal to:

(A) $3e^{\pi^2} + 5$

(B) $7e^{\pi^2} + 5$

(C) $2e^{\pi^2} + 5$

(D) $e^{\pi^2} + 5$

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(D) $e^{\pi^2} + 5$

Solution:

We have,

$$\frac{dy}{dx} = 2(y + 2 \sin x - 5)x - 2 \cos x$$

$$\Rightarrow \frac{dy}{dx} - 2xy = 2(2 \sin x - 5)x - 2 \cos x$$

$$I. F = e^{-x^2}$$

So,

$$y \cdot e^{-x^2} = \int e^{-x^2} (2x(2 \sin x - 5) - 2 \cos x) dx$$

$$\Rightarrow y \cdot e^{-x^2} = e^{-x^2} (5 - 2 \sin x) + c$$

$$\Rightarrow y = 5 - 2 \sin x + c \cdot e^{x^2}$$

$$\text{Given, } y(0) = 7$$

$$\Rightarrow 7 = 5 + c \Rightarrow c = 2$$

$$\therefore y = 5 - 2 \sin x + 2e^{x^2}$$

Now, $y(\pi)$ is

$$y = 5 + 2e^{\pi^2}$$

Q65: When a certain biased die is rolled, a particular face occurs with probability $\frac{1}{6} - x$ and its opposite face occurs with probability $\frac{1}{6} + x$. All other faces occur with probability $\frac{1}{6}$. Note that opposite faces sum to 7 in any die. If $0 < x < \frac{1}{6}$, and the probability of obtaining total sum = 7, when such a die is rolled twice, is $\frac{13}{96}$, then the value of x is:

- (A) $\frac{1}{9}$
- (B) $\frac{1}{12}$
- (C) $\frac{1}{8}$
- (D) $\frac{1}{16}$

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(A) $\frac{1}{9}$

(B) $\frac{1}{12}$

(C) $\frac{1}{8}$

(D) $\frac{1}{16}$

Solution:

Probability of obtaining total sum 7 = probability of getting the opposite face.

Probability of getting opposite faces

$$= 2 \left[\left(\frac{1}{6} - x \right) \left(\frac{1}{6} + x \right) + \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} \right]$$

Given,

$$\Rightarrow 2 \left[\left(\frac{1}{6} - x \right) \left(\frac{1}{6} + x \right) + \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} \right] = \frac{13}{96}$$

$$\Rightarrow \frac{6}{36} - 2x^2 = \frac{13}{96}$$

$$\Rightarrow x = \frac{1}{8}$$

Q66: A wire of length **20 m** is to be cut into two pieces. One of the pieces is to be made into a square and the other into a regular hexagon. Then the length of the side (in meters) of the hexagon, so that the combined area of the square and the hexagon is minimum, is:

(A) $\frac{10}{3+2\sqrt{3}}$

(B) $\frac{5}{3+\sqrt{3}}$

(C) $\frac{10}{2+3\sqrt{3}}$

(D) $\frac{5}{2+\sqrt{3}}$

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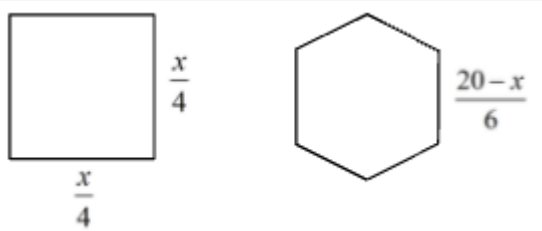
(A) $\frac{10}{3+2\sqrt{3}}$

(B) $\frac{5}{3+\sqrt{3}}$

(C) $\frac{10}{2+3\sqrt{3}}$

(D) $\frac{5}{2+\sqrt{3}}$

Solution:



Let the wire is cut into two pieces of length x and $20 - x$.

Area of square = $\left(\frac{x}{4}\right)^2$ and

Area of regular hexagon = $6 \times \frac{\sqrt{3}}{4} \left(\frac{20-x}{6}\right)^2$

Total area = $A(x) = \frac{x^2}{16} + \frac{3\sqrt{3}}{2} \frac{(20-x)^2}{36}$

$A'(x) = \frac{2x}{16} + \frac{3\sqrt{3} \times 2}{2 \times 36} (20-x)(-1)$

For the combined area to be minimum

$A'(x) = 0$

$\Rightarrow \frac{2x}{16} + \frac{3\sqrt{3} \times 2}{2 \times 36} (20-x)(-1) = 0$

$\Rightarrow x = \frac{40\sqrt{3}}{3+2\sqrt{3}}$

Length of side of regular Hexagon = $\frac{1}{6}(20-x)$
 $= \frac{1}{6} \left(20 - \frac{40\sqrt{3}}{3+2\sqrt{3}} \right)$
 $= \frac{10}{3+2\sqrt{3}} \text{ m}$

Q67: Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes $x - y - z - 1 = 0$ and $2x + y - 3z + 4 = 0$, is :

(A) $-x + 2y + 2z - 3 = 0$

(B) $3x - 4z + 3 = 0$

(C) $3x - y - 5z + 2 = 0$

(D) $4x - y - 5z + 2 = 0$

Q67: Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes $x - y - z - 1 = 0$ and $2x + y - 3z + 4 = 0$, is :

(A) $-x + 2y + 2z - 3 = 0$

(B) $3x - 4z + 3 = 0$

(C) $3x - y - 5z + 2 = 0$

(D) $4x - y - 5z + 2 = 0$

Solution:

Required equation of plane

$$P_1 + \lambda P_2 = 0$$

$$(x - y - z - 1) + \lambda (2x + y - 3z + 4) = 0$$

Given that its distance from origin is $\sqrt{\frac{2}{21}}$

$$\text{Thus } \frac{|4\lambda - 1|}{\sqrt{(2\lambda + 1)^2 + (\lambda - 1)^2 + (-3\lambda - 1)^2}} = \frac{\sqrt{2}}{\sqrt{21}}$$

$$\Rightarrow 21(4\lambda - 1)^2 = 2(14\lambda^2 + 8\lambda + 3)$$

$$\Rightarrow 336\lambda^2 - 168\lambda + 21 = 28\lambda^2 + 16\lambda + 6$$

$$\Rightarrow 308\lambda^2 - 184\lambda + 15 = 0$$

$$\Rightarrow 308\lambda^2 - 154\lambda - 30\lambda + 15 = 0$$

$$\Rightarrow (2\lambda - 1)(154\lambda - 15) = 0$$

$$\Rightarrow \lambda = \frac{1}{2} \text{ or } \frac{15}{154}$$

For $\lambda = \frac{1}{2}$, the required equation of plane is

$$4x - y - 5z + 2 = 0$$

Q68: The statement $(p \wedge (p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow r$ is :

- (A) equivalent to $q \rightarrow \sim r$
- (B) equivalent to $p \rightarrow \sim r$
- (C) a fallacy
- (D) a tautology

Q68: The statement $(p \wedge (p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow r$ is :

- (A) equivalent to $q \rightarrow \sim r$
- (B) equivalent to $p \rightarrow \sim r$
- (C) a fallacy
- (D) a tautology

Solution:

$$\begin{aligned} & (p \wedge (p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow r \\ &= (p \wedge (\sim p \vee q) \wedge (\sim q \vee r)) \rightarrow r \\ &= ((p \wedge q) \wedge (\sim q \vee r)) \rightarrow r \\ &= ((p \wedge q) \wedge r) \rightarrow r \\ &= \sim ((p \wedge q) \wedge r) \vee r \\ &= (\sim (p \wedge q) \vee \sim r) \vee r \\ &\Rightarrow \text{tautology} \end{aligned}$$

Q69: If for $x, y \in R, x > 0, y = (\log_{10} x) + \left(\log_{10} x^{\frac{1}{3}}\right) + \left(\log_{10} x^{\frac{1}{9}}\right) + \dots$ upto ∞ terms and $\frac{2+4+6+\dots+2y}{3+6+9+\dots+3y} = \frac{4}{\log_{10} x}$, then the ordered pair (x, y) is equal to :

- (A) $(10^6, 6)$
- (B) $(10^6, 9)$
- (C) $(10^2, 3)$
- (D) $(10^4, 6)$

Q69: If for $x, y \in R, x > 0, y = (\log_{10} x) + \left(\log_{10} x^{\frac{1}{3}}\right) + \left(\log_{10} x^{\frac{1}{9}}\right) + \dots$ upto ∞ terms and $\frac{2+4+6+\dots+2y}{3+6+9+\dots+3y} = \frac{4}{\log_{10} x}$, then the ordered pair (x, y) is equal to :

(A) $(10^6, 6)$

(B) $(10^6, 9)$

(C) $(10^2, 3)$

(D) $(10^4, 6)$

Solution:

We have,

$$\frac{2+4+6+\dots+2y}{3+6+9+\dots+3y} = \frac{4}{\log_{10} x},$$

$$\frac{2(1+2+3+\dots+y)}{3(1+2+3+\dots+y)} = \frac{4}{\log_{10} x}$$

$$\Rightarrow \log_{10} x = 6$$

$$\Rightarrow x = 10^6$$

Now,

$$y = (\log_{10} x) + \left(\log_{10} x^{\frac{1}{3}}\right) + \left(\log_{10} x^{\frac{1}{9}}\right) + \dots \infty$$

$$y = \left(1 + \frac{1}{3} + \frac{1}{9} + \dots \infty\right) \log_{10} x$$

$$y = \left(\frac{1}{1-\frac{1}{3}}\right) \log_{10} x$$

$$\therefore y = 9$$

$$\text{So, } (x, y) = (10^6, 9)$$

Q70: If $S = \{z \in \mathbb{C} : \frac{z-i}{z+2i} \in \mathbb{R}\}$, then:

- (A) S is a circle in the complex plane
- (B) S contains only one element
- (C) S is a straight line in the complex plane
- (D) S contains exactly two elements

Q70: If $S = \{z \in \mathbb{C} : \frac{z-i}{z+2i} \in \mathbb{R}\}$, then:

- (A) S is a circle in the complex plane
- (B) S contains only one element
- (C) S is a straight line in the complex plane
- (D) S contains exactly two elements

Solution:

Given, $\frac{z-i}{z+2i} \in \mathbb{R}$

$\Rightarrow \frac{z-i}{z+2i}$ is purely real

Let $z = x + iy$

$$\begin{aligned} \frac{x+iy-i}{x+iy+2i} &= \left(\frac{x+i(y-1)}{x+i(y+2)} \right) \left(\frac{x-i(y+2)}{x-i(y+2)} \right) \\ &= \frac{x^2+(y-1)(y+2)+i(-x(y+2)+x(y-1))}{x^2+(y+2)^2} = 0 \end{aligned}$$

$$\text{Imaginary part} = \frac{-x(y+2)+x(y-1)}{x^2+(y+2)^2} = 0$$

$$\Rightarrow -xy - 2x + xy - x = 0$$

$$\Rightarrow -3x = 0$$

$$\therefore x = 0$$

$\Rightarrow S$ is a straight line in the complex plane.

Q71: $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to:

(A) 10

(B) 8

(C) 6

(D) 5

Q71: $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to:

(A) 10

(B) 8

(C) 6

(D) 5

Solution:

$$\text{Let } I = \int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$$

$$I = \int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x - 22)^2} dx \dots (1)$$

We know,

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$$

$$\text{So } I = \int_6^{16} \frac{\log_e (22 - x)^2}{\log_e (22 - x)^2 + \log_e (22 - (22 - x))^2} dx$$

$$I = \int_6^{16} \frac{\log_e (22 - x)^2}{\log_e x^2 + \log_e (22 - x)^2} dx \dots (2)$$

$$(1) + (2)$$

$$2I = \int_6^{16} 1 \cdot dx$$

$$2I = 16 - 6 = 10$$

$$\therefore I = 5$$

Q72: If $0 < x < 1$, then $\frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots$, is equal to:

(A) $x \left(\frac{1+x}{1-x} \right) + \log_e (1-x)$

(B) $x \left(\frac{1-x}{1+x} \right) + \log_e (1-x)$

(C) $\frac{1-x}{1+x} + \log_e (1-x)$

(D) $\frac{1+x}{1-x} + \log_e (1-x)$

Q72: If $0 < x < 1$, then $\frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots$, is equal to:

(A) $x \left(\frac{1+x}{1-x} \right) + \log_e (1-x)$

(B) $x \left(\frac{1-x}{1+x} \right) + \log_e (1-x)$

(C) $\frac{1-x}{1+x} + \log_e (1-x)$

(D) $\frac{1+x}{1-x} + \log_e (1-x)$

Solution:

$$\begin{aligned}\text{Let } t &= \frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots \infty \\ &= \left(2 - \frac{1}{2}\right)x^2 + \left(2 - \frac{1}{3}\right)x^3 + \left(2 - \frac{1}{4}\right)x^4 + \dots \infty \\ &= 2(x^2 + x^3 + x^4 + \dots \infty) - \left(\frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \infty\right) \\ &= \frac{2x^2}{1-x} - (-\log_e (1-x) - x) \\ \Rightarrow t &= \frac{2x^2}{1-x} + x + \log_e (1-x) \\ \Rightarrow t &= \frac{x(1+x)}{1-x} + \log_e (1-x)\end{aligned}$$

Q73: The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are $2, 3, -6$ is:

- (A) 2
- (B) 5
- (C) 3
- (D) 1

Q73: The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are $2, 3, -6$ is:

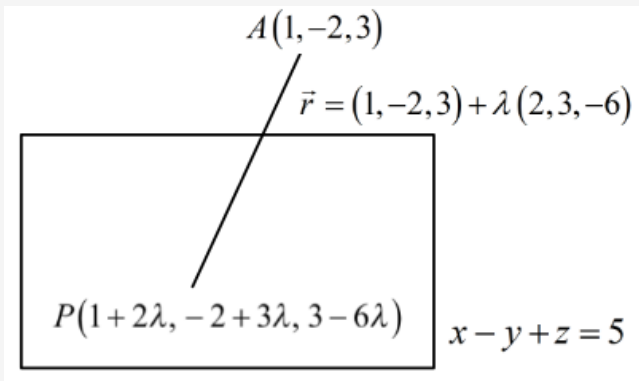
(A) 2

(B) 5

(C) 3

(D) 1

Solution:



Now,

$$x - y + z = 5$$

$$\Rightarrow (1 + 2\lambda) + 2 - 3\lambda + 3 - 6\lambda = 5$$

$$\Rightarrow 6 - 7\lambda = 5 \Rightarrow \lambda = \frac{1}{7}$$

$$P = \left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$$

$$AP = \sqrt{\left(1 - \frac{9}{7}\right)^2 + \left(-2 + \frac{11}{7}\right)^2 + \left(3 - \frac{15}{7}\right)^2}$$

$$AP = \sqrt{\left(\frac{4}{49}\right) + \frac{9}{49} + \frac{36}{49}}$$

$$\therefore AP = 1$$

Q74: If $x^2 + 9y^2 - 4x + 3 = 0, x, y \in R$, then x and y respectively lie in the intervals:

- (A) $[1, 3]$ and $[-\frac{1}{3}, \frac{1}{3}]$
- (B) $[-\frac{1}{3}, \frac{1}{3}]$ and $[1, 3]$
- (C) $[1, 3]$ and $[1, 3]$
- (D) $[-\frac{1}{3}, \frac{1}{3}]$ and $[-\frac{1}{3}, \frac{1}{3}]$

Q74: If $x^2 + 9y^2 - 4x + 3 = 0$, $x, y \in R$, then x and y respectively lie in the intervals:

- (A) $[1, 3]$ and $[-\frac{1}{3}, \frac{1}{3}]$
- (B) $[-\frac{1}{3}, \frac{1}{3}]$ and $[1, 3]$
- (C) $[1, 3]$ and $[1, 3]$
- (D) $[-\frac{1}{3}, \frac{1}{3}]$ and $[-\frac{1}{3}, \frac{1}{3}]$

Solution:

$$x^2 + 9y^2 - 4x + 3 = 0$$

$$(x^2 - 4x) + (9y^2) + 3 = 0$$

$$(x^2 - 4x + 4) + (9y^2) + 3 - 4 = 0$$

$$(x - 2)^2 + (3y)^2 = 1$$

$$\frac{(x-2)^2}{(1)^2} + \frac{y^2}{(\frac{1}{3})^2} = 1 \text{ (equation of an ellipse).}$$

As it is an equation of an ellipse, x and y can vary inside the ellipse.

So, $x - 2 \in [-1, 1]$ and $y \in [-\frac{1}{3}, \frac{1}{3}]$

$x \in [1, 3]$ and $y \in [-\frac{1}{3}, \frac{1}{3}]$

Q75: Let $\frac{\sin A}{\sin B} = \frac{\sin(A-C)}{\sin(C-B)}$, where A, B, C are angles of a triangle ABC . If the lengths of the sides opposite to these angles are a, b, c respectively, then:

- (A) c^2, a^2, b^2 in A.P.
- (B) $b^2 - a^2 = a^2 + c^2$
- (C) b^2, c^2, a^2 are in A.P.
- (D) a^2, b^2, c^2 are in A.P.

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(D) a^2, b^2, c^2 are in A.P.

Solution:

$$\frac{\sin A}{\sin B} = \frac{\sin(A-C)}{\sin(C-B)}$$

Since, A, B, C are angles of triangle ABC .

$$A + B + C = \pi$$

$$A = \pi - (B + C)$$

$$\text{So, } \sin A = \sin(B + C) \dots (1)$$

$$\text{Similarly, } \sin B = \sin(A + C) \dots (2)$$

From (1) and (2)

$$\frac{\sin A}{\sin B} = \frac{\sin(A-C)}{\sin(C-B)}$$

$$\Rightarrow \sin(C + B) \cdot \sin(C - B) = \sin(A - C) \sin(A + C)$$

$$\Rightarrow \sin^2 C - \sin^2 B = \sin^2 A - \sin^2 C \quad (\because \sin(x + y) \sin(x - y) = \sin^2 x - \sin^2 y)$$

$$\Rightarrow 2\sin^2 C = \sin^2 A + \sin^2 B$$

By sine rule

$$2c^2 = a^2 + b^2$$

$$\Rightarrow b^2, c^2 \text{ and } a^2 \text{ are in A.P.}$$

Q76: $\sum_{k=0}^{20} \left({}^{20}C_k\right)^2$ is equal to:

(A) ${}^{41}C_{20}$

(B) ${}^{40}C_{20}$

(C) ${}^{40}C_{21}$

(D) ${}^{40}C_{19}$

Q76: $\sum_{k=0}^{20} \left({}^{20}C_k\right)^2$ is equal to:

(A) ${}^{41}C_{20}$

(B) ${}^{40}C_{20}$

(C) ${}^{40}C_{21}$

(D) ${}^{40}C_{19}$

Solution:

$$\begin{aligned} & \sum_{k=0}^{20} \left({}^{20}C_k\right)^2 \\ &= \left({}^{20}C_0\right)^2 + \left({}^{20}C_1\right)^2 + \dots + \left({}^{20}C_{20}\right)^2 \\ &= {}^{20}C_0 \cdot {}^{20}C_{20} + {}^{20}C_1 \cdot {}^{20}C_{19} + \dots + {}^{20}C_{20} \cdot {}^{20}C_0 \\ &\text{Since } {}^{20}C_k \text{ is the coefficient of } x^k \text{ in } (1+x)^{20} \\ &= \text{coefficient of } x^{20} \text{ in } (1+x)^{40} \\ &= {}^{40}C_{20} \end{aligned}$$

Q77: If $U_n = \left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right)^2 \dots \left(1 + \frac{n^2}{n^2}\right)^n$, then $\lim_{n \rightarrow \infty} (U_n)^{\frac{-4}{n^2}}$ is equal to:

(A) $\frac{4}{e^2}$

(B) $\frac{4}{e}$

(C) $\frac{16}{e^2}$

(D) $\frac{e^2}{16}$

Q77: If $U_n = \left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right)^2 \dots \left(1 + \frac{n^2}{n^2}\right)^n$, then $\lim_{n \rightarrow \infty} (U_n)^{\frac{-4}{n^2}}$ is equal to:

(A) $\frac{4}{e^2}$

(B) $\frac{4}{e}$

(C) $\frac{16}{e^2}$

(D) $\frac{e^2}{16}$

Solution:

$$U_n = \prod_{r=1}^n \left(1 + \frac{r^2}{n^2}\right)^r$$

$$L = \lim_{n \rightarrow \infty} (U_n)^{-\frac{4}{n^2}}$$

$$\log L = \lim_{n \rightarrow \infty} \sum_{r=1}^n -\frac{4}{n^2} \log \left(1 + \frac{r^2}{n^2}\right)^r$$

$$\Rightarrow \log L = \lim_{n \rightarrow \infty} \sum_{r=1}^n -\frac{4r}{n} \cdot \frac{1}{n} \log \left(1 + \frac{r^2}{n^2}\right)$$

$$\Rightarrow \log L = -4 \int_0^1 (x \log(1 + x^2)) dx$$

$$\text{put } 1 + x^2 = t$$

$$\text{Now, } 2x dx = dt$$

$$= -2 \int_1^2 \log(t) dt = -2[t \log t - t]_1^2$$

$$\Rightarrow \log L = -2(2 \log 2 - 1)$$

$$\therefore L = e^{-2(2 \log 2 - 1)}$$

$$= e^{-2\left(\log\left(\frac{4}{e}\right)\right)}$$

$$= e^{\log\left(\frac{4}{e}\right)^{-2}}$$

$$= \left(\frac{e}{4}\right)^2 = \frac{e^2}{16}$$

$$\therefore \lim_{n \rightarrow \infty} (U_n)^{-\frac{4}{n^2}} = \frac{e^2}{16}$$

Q78: If the matrix $A = \begin{pmatrix} 0 & 2 \\ k & -1 \end{pmatrix}$ satisfies $A(A^3 + 3I) = 2I$, then the value of k is:

(A) 1

(B) -1

(C) $-\frac{1}{2}$

(D) $\frac{1}{2}$

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(A) 1

(B) -1

(C) $-\frac{1}{2}$

(D) $\frac{1}{2}$

Solution:

Given matrix $A = \begin{bmatrix} 0 & 2 \\ k & -1 \end{bmatrix}$ and also

$$A(A^3 + 3I) = 2I,$$

$$A^4 + 3IA = 2I$$

$$\begin{aligned} A^2 &= \begin{bmatrix} 0 & 2 \\ k & -1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ k & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2k & -2 \\ -k & 2k+1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A^4 &= (A^2)^2 = \begin{bmatrix} 2k & -2 \\ -k & 2k+1 \end{bmatrix} \begin{bmatrix} 2k & -2 \\ -k & 2k+1 \end{bmatrix} \\ &= \begin{bmatrix} 4k^2 + 2k & -8k - 2 \\ -4k^2 - k & 2k + (2k+1)^2 \end{bmatrix} \\ &= \begin{bmatrix} 4k^2 + 2k & -8k - 2 \\ -4k^2 - k & 4k^2 + 6k + 1 \end{bmatrix} \end{aligned}$$

$$A^4 + 3A = \begin{bmatrix} 4k^2 + 2k & -8k + 4 \\ -4k^2 + 2k & 4k^2 + 6k - 2 \end{bmatrix} = 2I$$

$$\Rightarrow \begin{bmatrix} 4k^2 + 2k & -8k + 4 \\ -4k^2 + 2k & 4k^2 + 6k - 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\Rightarrow -8k + 4 = 0$$

$$\Rightarrow k = \frac{1}{2}$$

Q79: Let us consider a curve, $y = f(x)$ passing through the point $(-2, 2)$ and the slope of the tangent to the curve at any point $(x, f(x))$ is given by $f(x) + xf'(x) = x^2$. Then:

(A) $x^2 + 2xf(x) + 4 = 0$

(B) $x^3 - 3xf(x) - 4 = 0$

(C) $x^3 + xf(x) + 12 = 0$

(D) $x^2 + 2xf(x) - 12 = 0$

Q79: Let us consider a curve, $y = f(x)$ passing through the point $(-2, 2)$ and the slope of the tangent to the curve at any point $(x, f(x))$ is given by $f(x) + xf'(x) = x^2$. Then:

(A) $x^2 + 2xf(x) + 4 = 0$

(B) $x^3 - 3xf(x) - 4 = 0$

(C) $x^3 + xf(x) + 12 = 0$

(D) $x^2 + 2xf(x) - 12 = 0$

Solution:

$$y + x \frac{dy}{dx} = x^2 \text{ (given)}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = x$$

$$I.F = e^{\int \frac{1}{x} dx} = x$$

Now, the solution differential equation is

$$\Rightarrow y \cdot x = \int x \cdot x \cdot dx$$

$$\Rightarrow y \cdot x = \int x^2 \cdot dx$$

$$\Rightarrow xy = \frac{x^3}{3} + c$$

It passes through $(-2, 2)$, so

$$-4 = -\frac{8}{3} + c$$

$$c = -\frac{4}{3}$$

$$\therefore 3xy = x^3 - 4$$

That is $3xf(x) = x^3 - 4$

Q80: If $(\sin^{-1} x)^2 - (\cos^{-1} x)^2 = a$; $0 < x < 1$, $a \neq 0$, then value of $2x^2 - 1$ is:

(A) $\cos\left(\frac{2a}{\pi}\right)$

(B) $\sin\left(\frac{4a}{\pi}\right)$

(C) $\cos\left(\frac{4a}{\pi}\right)$

(D) $\sin\left(\frac{2a}{\pi}\right)$

Q80: If $(\sin^{-1} x)^2 - (\cos^{-1} x)^2 = a$; $0 < x < 1$, $a \neq 0$, then value of $2x^2 - 1$ is:

(A) $\cos\left(\frac{2a}{\pi}\right)$

(B) $\sin\left(\frac{4a}{\pi}\right)$

(C) $\cos\left(\frac{4a}{\pi}\right)$

(D) $\sin\left(\frac{2a}{\pi}\right)$

Solution:

Given

$$a = (\sin^{-1} x)^2 - (\cos^{-1} x)^2$$

$$a = (\sin^{-1} x + \cos^{-1} x) (\sin^{-1} x - \cos^{-1} x)$$

$$a = \frac{\pi}{2} \left(\frac{\pi}{2} - 2\cos^{-1} x \right)$$

$$\Rightarrow 2\cos^{-1} x = \frac{\pi}{2} - \frac{2a}{\pi}$$

$$\Rightarrow \cos^{-1} (2x^2 - 1) = \frac{\pi}{2} - \frac{2a}{\pi}$$

$$\Rightarrow 2x^2 - 1 = \cos\left(\frac{\pi}{2} - \frac{2a}{\pi}\right)$$

$$\therefore 2x^2 - 1 = \sin\left(\frac{2a}{\pi}\right)$$

Q81: Let $\vec{a} = \hat{i} + 5\hat{j} + \alpha\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $|\vec{b} \times \vec{c}| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $|\vec{a}|^2$ is _____.

Q81: Let $\vec{a} = \hat{i} + 5\hat{j} + \alpha\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $|\vec{b} \times \vec{c}| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $|\vec{a}|^2$ is _____.

90.00

Solution:

Since $\vec{a} \cdot \vec{b} = 0$

$$1 + 15 + \alpha\beta = 0 \Rightarrow \alpha\beta = -16$$

Also,

$$|\vec{b} \times \vec{c}|^2 = 75$$

$$\Rightarrow (9 + 2\beta)^2 + (3 - \beta)^2 + (5)^2 = 75$$

$$\Rightarrow 5\beta^2 + 30\beta + 40 = 0$$

$$\Rightarrow \beta = -4, -2$$

$$\Rightarrow \alpha = 4, 8$$

$$\Rightarrow |\vec{a}|_{\max}^2 = (26 + \alpha^2)_{\max} = 90$$

Q82: The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is

 .

Q82: The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is

4.00

Solution:

Given,

$$3x^4 + 4x^3 - 12x^2 + 4 = 0$$

So, Let $f(x) = 3x^4 + 4x^3 - 12x^2 + 4$

$$\therefore f'(x) = 12x(x^2 + x - 2)$$

$$\Rightarrow f'(x) = 12x(x + 2)(x - 1)$$

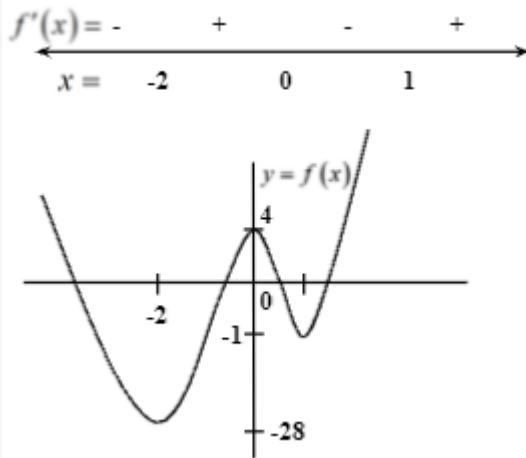
$$f'(x) = 0$$

$$\Rightarrow x = -2, 0, 1,$$

$$f(0) = 4 > 0$$

$$f(-2) = -28 < 0$$

$$f(1) = -1 < 0$$



Hence, $f(x)$ has 4 distinct real roots.

Q83: Let n be an odd natural number such that the variance of $1, 2, 3, 4, \dots, n$ is 14, then n is equal to _____.

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13.00

Solution:

$$\text{Variance} = \frac{\sum x^2}{n} - \frac{(\sum x)^2}{n^2}$$

$$14 = \frac{1^2+2^2+\dots+n^2}{n} - \frac{(1+2+3+\dots+n)^2}{n^2}$$

$$\Rightarrow 14 = \frac{n(n+1)(2n+1)}{6n} - \left(\frac{n(n+1)}{2n}\right)^2$$

$$\Rightarrow 14 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$\Rightarrow 28 = \frac{2n^2+3n+1}{3} - \frac{n^2+2n+1}{2}$$

$$\Rightarrow 28 = \frac{4n^2+6n+2-3n^2-6n-3}{6}$$

$$\Rightarrow 28 = \frac{n^2-1}{6}$$

$$\Rightarrow n^2 - 1 = 168$$

$$\Rightarrow n^2 = 169$$

$$\Rightarrow n = 13$$

Q84: If the system of linear equations

$$2x + y - z = 3$$

$$x - y - z = \alpha$$

$$3x + 3y + \beta z = 3$$

has infinitely many solutions, then $\alpha + \beta - \alpha\beta$ is equal to _____.

Q84: If the system of linear equations

$$2x + y - z = 3$$

$$x - y - z = \alpha$$

$$3x + 3y + \beta z = 3$$

has infinitely many solutions, then $\alpha + \beta - \alpha\beta$ is equal to _____.

Solution:

$$2x + y - z = 3 \quad \dots (i)$$

$$x - y - z = \alpha \quad \dots (ii)$$

$$3x + 3y + \beta z = 3 \quad (iii)$$

Now,

$2 \times (i) - (ii) - (iii)$ gives:

$$-(1 + \beta)z = 3 - \alpha$$

For infinitely many solutions.

$$\beta + 1 = 0 = 3 - \alpha \Rightarrow (\alpha, \beta) = (3, -1)$$

$$\text{Hence, } \alpha + \beta - \alpha\beta = 5$$

Q85: If $A = \{x \in R : |x - 2| > 1\}$, $B = \{x \in R : \sqrt{x^2 - 3} > 1\}$,
 $C = \{x \in R : |x - 4| \geq 2\}$ and Z is the set of all integers, then the number of subsets
of the set $(A \cap B \cap C)^c \cap Z$ is _____.

Q85: If $A = \{x \in R : |x - 2| > 1\}$, $B = \{x \in R : \sqrt{x^2 - 3} > 1\}$,
 $C = \{x \in R : |x - 4| \geq 2\}$ and Z is the set of all integers, then the number of subsets
of the set $(A \cap B \cap C)^c \cap Z$ is _____.

256.00

Solution:

$$A = \{x \in R : |x - 2| > 1\},$$

$$A = x \in (-\infty, 1) \cup (3, \infty)$$

$$B = \{x \in R : \sqrt{x^2 - 3} > 1\},$$

$$B = x \in (-\infty, -2) \cup (2, \infty)$$

$$C = \{x \in R : |x - 4| \geq 2\}$$

$$C = x \in (-\infty, 2] \cup [6, \infty)$$

$$\text{So, } A \cap B \cap C = (-\infty, -2) \cup [6, \infty)$$

$$(A \cap B \cap C)^c = [-2, 6)$$

$$(A \cap B \cap C)^c \cap Z = \{-2, -1, 0, 1, 2, 3, 4, 5\}$$

$$\text{Hence, the number of subsets} = 2^8 = 256$$

Q86: If the minimum area of the triangle formed by a tangent to the ellipse $\frac{x^2}{b^2} + \frac{y^2}{4a^2} = 1$ and the coordinate axis is kab , then k is equal to _____.

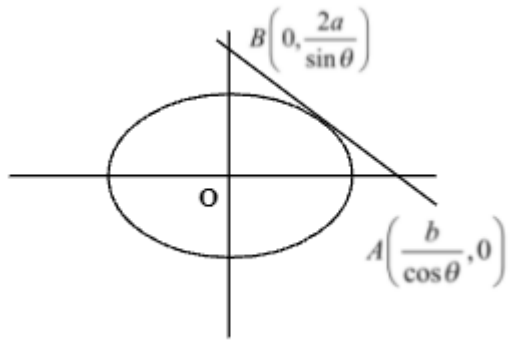
Q86: If the minimum area of the triangle formed by a tangent to the ellipse $\frac{x^2}{b^2} + \frac{y^2}{4a^2} = 1$ and the coordinate axis is kab , then k is equal to _____.

2.00

Solution:

Equation of tangent at $(b \cos \theta, 2a \sin \theta)$ is

$$\frac{x \cos \theta}{b} + \frac{y \sin \theta}{2a} = 1$$



$$\begin{aligned} \text{So, Area } (\Delta OAB) &= \frac{1}{2} \times OB \times OA \\ &= \frac{1}{2} \times \frac{b}{\cos \theta} \times \frac{2a}{\sin \theta} \\ &= \frac{2ab}{\sin 2\theta} \geq 2ab \end{aligned}$$

$$\begin{aligned} \therefore \text{The minimum area} &= 2ab \\ \Rightarrow k &= 2 \end{aligned}$$

Q87: A number is called a palindrome if it reads the same backward as well as forward. For example 285582 is a six digit palindrome. The number of six digit palindromes, which are divisible by 55, is_____.

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100.00

Solution:

5	a	b	b	a	5
---	---	---	---	---	---

It is always divisible by **5** and **11**.

b can be selected in 10 ways and **a** can be selected in 10 ways.

So, the required number = $10 \times 10 = 100$

Q88: If $\int \frac{dx}{(x^2+x+1)^2} = a \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + b \left(\frac{2x+1}{x^2+x+1} \right) + C, x > 0$ where C is the constant of integration, then the value of $9(\sqrt{3}a + b)$ is equal to _____.

Q88: If $\int \frac{dx}{(x^2+x+1)^2} = a \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + b \left(\frac{2x+1}{x^2+x+1} \right) + C, x > 0$ where C is the constant of integration, then the value of $9(\sqrt{3}a + b)$ is equal to _____.

Solution:

$$\text{Let } I = \int \frac{dx}{(x^2+x+1)^2}$$

$$I = \int \frac{dx}{\left[\left(x+\frac{1}{2}\right)^2 + \frac{3}{4}\right]^2}$$

$$\text{Put } x + \frac{1}{2} = t$$

$$\Rightarrow dx = dt$$

$$= \int \frac{dt}{\left(t^2 + \frac{3}{4}\right)^2}$$

$$\text{Put } t = \frac{\sqrt{3}}{2} \tan \theta$$

$$\Rightarrow dt = \frac{\sqrt{3}}{2} \sec^2 \theta d\theta$$

$$\therefore I = \frac{\sqrt{3}}{2} \int \frac{\sec^2 \theta d\theta}{\frac{9}{16} \sec^4 \theta}$$

$$= \frac{4\sqrt{3}}{9} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{4\sqrt{3}}{9} \left(\theta + \frac{\sin 2\theta}{2} \right) + C$$

$$= \frac{4\sqrt{3}}{9} \left[\tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + \frac{\sqrt{3}(2x+1)}{3+(2x+1)^2} \right] + C$$

$$= \frac{4\sqrt{3}}{9} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + \frac{1}{3} \left(\frac{2x+1}{x^2+x+1} \right) + C$$

$$\text{Hence, } 9(\sqrt{3}a + b) = 15$$

Q89: If $y^{\frac{1}{4}} + y^{-\frac{1}{4}} = 2x$, and $(x^2 - 1) \frac{d^2y}{dx^2} + \alpha x \frac{dy}{dx} + \beta y = 0$, then $|\alpha - \beta|$ is equal to

_____.

Q89: If $y^{\frac{1}{4}} + y^{-\frac{1}{4}} = 2x$, and $(x^2 - 1) \frac{d^2y}{dx^2} + \alpha x \frac{dy}{dx} + \beta y = 0$, then $|\alpha - \beta|$ is equal to

_____.

17.00

Solution:

$$y^{\frac{1}{4}} + y^{-\frac{1}{4}} = 2x$$

$$\Rightarrow y^{\frac{1}{4}} + \frac{1}{y^{\frac{1}{4}}} = 2x$$

$$\Rightarrow \left(y^{\frac{1}{4}}\right)^2 - 2xy^{\left(\frac{1}{4}\right)} + 1 = 0$$

$$\Rightarrow y^{\frac{1}{4}} = x + \sqrt{x^2 - 1} \text{ or } x - \sqrt{x^2 - 1}$$

$$\text{So, } y^{\frac{1}{4}} = x + \sqrt{x^2 - 1}$$

$$\frac{1}{4} \frac{1}{y^{\frac{3}{4}}} \frac{dy}{dx} = 1 + \frac{x}{\sqrt{x^2 - 1}}$$

$$\Rightarrow \frac{1}{4} \frac{1}{y^{\frac{3}{4}}} \frac{dy}{dx} = \frac{y^{\frac{1}{4}}}{\sqrt{x^2 - 1}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{4y}{\sqrt{x^2 - 1}} \dots (1)$$

$$\text{Hence, } \frac{d^2y}{dx^2} = 4 \frac{(\sqrt{x^2 - 1})y' - \frac{yx}{\sqrt{x^2 - 1}}}{(x^2 - 1)}$$

$$\Rightarrow (x^2 - 1) y'' = 4 \left(\sqrt{x^2 - 1} y' - \frac{xy}{\sqrt{x^2 - 1}} \right)$$

$$\Rightarrow (x^2 - 1) y'' = 4 \left(4y - \frac{xy'}{4} \right) \text{ from (1)}$$

$$\Rightarrow (x^2 - 1) y'' + xy' - 16y = 0$$

$$\therefore (x^2 - 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - 16y = 0,$$

$$\text{So, } |\alpha - \beta| = |1 + 16| = 17$$

Q90: Let the equation $x^2 + y^2 + px + (1 - p)y + 5 = 0$ represent circles of varying radius $r \in (0, 5]$. Then the number of elements in the set $S = \{q : q = p^2 \text{ and } q \text{ is an integer}\}$ is _____.

Q90: Let the equation $x^2 + y^2 + px + (1 - p)y + 5 = 0$ represent circles of varying radius $r \in (0, 5]$. Then the number of elements in the set $S = \{q : q = p^2 \text{ and } q \text{ is an integer}\}$ is _____.

61.00

Solution:

$$r = \sqrt{\frac{p^2}{4} + \frac{(1-p)^2}{4}} - 5$$

$$r = \frac{\sqrt{2p^2 - 2p - 19}}{2}$$

Since, $r \in (0, 5]$

So, $0 < 2p^2 - 2p - 19 \leq 100$

$$\Rightarrow p \in \left[\frac{1-\sqrt{239}}{2}, \frac{1+\sqrt{39}}{2} \right) \cup \left(\frac{1+\sqrt{39}}{2}, \frac{1+\sqrt{239}}{2} \right]$$

So, number of integral values of p^2 is **61**.